Prova 1

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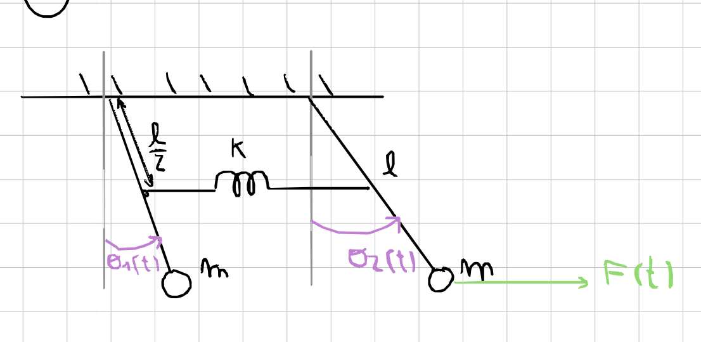

$x_1 = \theta_1 \quad x_2 = \dot\theta_1 \quad x_3 = \theta_2 \quad x_4 = \dot\theta_2$

Equazione di Lagrange: non ci sono forze d'attrito

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot q_i}\right) - \frac{\partial T}{\partial q_i} + \frac{\partial U}{\partial q_i} + \frac{\partial R}{\partial \dot q_i} = Q_i$$

- $T = T_1 + T_2 \qquad T_1 = \frac{1}{2} m v_G^2$ (no ω)

$v_G = \dot{OG} = \frac{d}{dt}\left(\langle l\sin\theta_1, l\cos\theta_1\rangle\right) \Rightarrow \langle l\dot\theta_1\cos\theta_1, -l\dot\theta_1\sin\theta_1\rangle$

$v_G^2 = \left(\sqrt{l^2\dot\theta_1^2\cancel{\cos^2\theta_1} + l^2\dot\theta_1^2\cancel{\sin^2\theta_1}}\right)^2 = l^2\dot\theta_1^2 \Rightarrow T_1 = \frac{1}{2} m l^2\dot\theta_1^2$

$T_2 = \frac{1}{2} m l^2\dot\theta_2^2$

$T = \frac{1}{2} m l^2(\dot\theta_1^2 + \dot\theta_2^2)$

- $U = U_p + U_k$

$U_p = U_{p1} + U_{p2} = -m g l\cos\theta_1 - m g l\cos\theta_2 = -mgl(\cos\theta_1 + \cos\theta_2)$

$U_k = \frac{1}{2} K s^2(t) = \frac{1}{2} k\left(\frac{l}{2}\sin\theta_2 - \frac{l}{2}\sin\theta_1\right)^2 = \frac{l^2 k}{8}(\sin\theta_2 - \sin\theta_1)^2$

$\hookrightarrow = \frac{l^2 k}{8}(\sin^2\theta_2 + \sin^2\theta_1 - 2\sin\theta_1\sin\theta_2)$

$\Rightarrow U = mgl(\cos\theta_1 + \cos\theta_2) + \frac{l^2 k}{8}(\sin\theta_2 - \sin\theta_1)^2$

Per θ_1:

RICORDA CHE $(\sin(x_1))^2 = 2\sin x_1\cdot\cos(x_1)$

- $\frac{d}{dt}\left(\frac{\partial T}{\partial\dot\theta_1}\right) = \frac{d}{dt}(m l^2\dot\theta_1) \Rightarrow m l^2\ddot\theta_1$
- $\frac{\partial T}{\partial\theta_1} = 0$
- $\frac{\partial U}{\partial\theta_1} = \frac{l^2 K}{8}(2\sin\theta_1\cos\theta_1 - 2\cos\theta_1\sin\theta_2) + mgl\sin\theta_1$

Per θ_2:

- $\frac{d}{dt}\left(\frac{\partial T}{\partial\dot\theta_1}\right) = \frac{d}{dt}(m l^2\dot\theta_2) \Rightarrow m l^2\ddot\theta_2$
- $\frac{\partial T}{\partial\theta_2} =$
- $\frac{\partial U}{\partial\theta_2} = \frac{l^2 K}{8}(2\sin\theta_2\cos\theta_2 - 2\cos\theta_2\sin\theta_1) + mgl\sin\theta_2$

- Calcolo delle Q_i

$F(t)\, l\cos\theta_2\cdot\delta\theta_2 = Q_1\cdot\delta\theta_1 + Q_2\cdot\delta\theta_2$

comparando $\Rightarrow Q_1 = 0 \quad Q_2 = F(t)\, l\cos\theta_2$

$\delta L = \sum F_i\cdot\delta r_i = \sum F_i\sum\frac{\delta r_i}{\partial q_k}\cdot\delta q_k = \sum Q_k\cdot\delta q_k$

$\delta x_2 = l\cdot\cos(\theta_2)$

$\delta L = F(t)\cdot\delta x_2 = F(t)\cdot\frac{\delta x_2}{\partial\theta_1}\cdot\delta\theta_1 + \frac{\delta x_2}{\partial\theta_2}\cdot\delta\theta_2 =$

- Equazioni

$$\begin{cases} m l^2\ddot\theta_1 + \frac{l^2 K}{8}(2\sin\theta_1\cos\theta_1 - 2\cos\theta_1\sin\theta_2) + mgl\sin\theta_1 = 0 \\ m l^2\ddot\theta_2 + \frac{l^2 K}{8}(2\sin\theta_2\cos\theta_2 - 2\cos\theta_2\sin\theta_1) + mgl\sin\theta_2 = F(t)\cdot l\cos\theta_2 \end{cases}$$

$$\Rightarrow \begin{cases} m l^2\dot x_2 + \frac{l^2 K}{8}(2\sin\theta_1\cos\theta_1 - 2\cos\theta_1\sin\theta_2) + mgl\sin x_1 = 0 \\ m l^2\dot x_4 + \frac{l^2 K}{8}(2\sin\theta_2\cos\theta_2 - 2\cos\theta_2\sin\theta_1) + mgl\sin x_3 = u\cdot l\cos x_3 \end{cases}$$

$$\begin{cases} \dot x_1 = x_2 \\ \dot x_2 = -\frac{K}{8m}(2\sin x_1\cos x_1 - 2\cos x_1\sin x_3) - \frac{g}{l}\sin x_1 \\ \dot x_3 = x_4 \\ \dot x_4 = -\frac{K}{8m}(2\sin x_3\cos x_3 - 2\cos x_3\sin x_1) - \frac{g}{l}\sin x_3 + \frac{u}{ml}\cos x_3 \end{cases}$$

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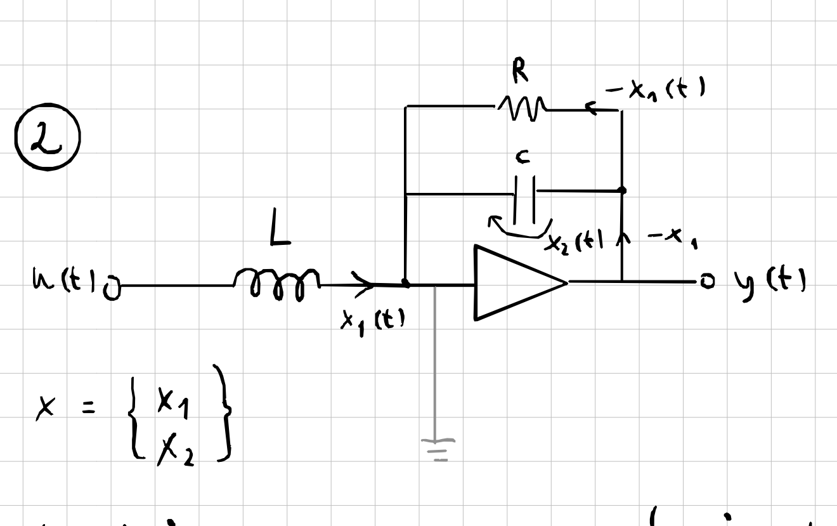


$x = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}$

$u - L\dot x_1 = 0$

$-x_1 - C\dot x_2 - \frac{x_2}{R} = 0$

$\Rightarrow \begin{cases} \dot x_1 = \frac{u}{L} \\ \dot x_2 = \frac{x_1}{C} - \frac{x_2}{RC} \end{cases}$

$A = \begin{bmatrix} 0 & 0 \\ \frac{1}{C} & \frac{-1}{RC} \end{bmatrix} \qquad B = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix}$

$y(t) = x_2 \Rightarrow C = [0 \;\; 1] \quad D = 0$

- $L = 1 \quad R = 100 \quad C = 0.005$

Calcolare $y_l(t)$ a partire da $x_0 = \begin{bmatrix} 0.1 \\ 1 \end{bmatrix}$

$A = \begin{bmatrix} 0 & 0 \\ \frac{1}{0.005} & \frac{-1}{0.5} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 200 & -2 \end{bmatrix} \qquad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$y_L(t) = C x(t) = C\cdot e^{At}\cdot x_0$

$e^{At} = \mathcal{L}^{-1}[(sI - A)^{-1}]$

$sI - A = \begin{bmatrix} s & 0 \\ 200 & s+2 \end{bmatrix} \quad \det(sI - A) = s^2 + 2s \Rightarrow s_1 = -2 \quad s_2 = 0$

$\Rightarrow$ modi: e^0, e^{-2t}

$y_L(t)$ è nella forma $y_L = K_1\cdot e^{-2t} + K_2\cdot e^0$

Mettiamo a sistema:

$\begin{cases} y_L(0) \\ \dot y_L(0) \end{cases}$

$\dot y_L = \dot x_2 = -\frac{x_1}{C} - \frac{x_2}{RC} = \frac{-x_1}{0.005} - \frac{x_2}{0.5}$

$$\begin{cases} y_L(0) = K_1\cdot 1 + K_2\cdot 1 \\ y_L(0) = x_2(0) = 1 \\ \dot y_L(0) = -\frac{0.1}{0.005} - \frac{1}{0.5} = -22 \\ \dot y_L(0) = \frac{d}{dt}(K_1 e^{-2t} + K_2\cdot e^0) = -2K_1 \end{cases}$$

$\Rightarrow \begin{cases} K_1 + K_2 = 1 \\ -2K_1 = -22 \end{cases} \Rightarrow \begin{cases} K_2 = -10 \\ K_1 = 11 \end{cases} \Rightarrow y_L(t) = 11e^{-2t} - 10$

- Modulo e argomento della risposta armonica

$(sI - A)^{-1} = \begin{bmatrix} \frac{1}{s} & 0 \\ \frac{200}{s(s+2)} & \frac{1}{s+2} \end{bmatrix} \qquad C = [0 \;\; 1] \qquad B = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix}$

$G(s) = C\cdot(sI - A)^{-1}\cdot B \Rightarrow [0 \;\; 1]\begin{bmatrix} \frac{1}{s} & 0 \\ \frac{200}{s(s+2)} & \frac{1}{s+2} \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\Rightarrow [0 \;\; 1]\begin{bmatrix} \frac{1}{s} & 0 \\ \frac{200}{s(s+2)} & 0 \end{bmatrix} = \frac{200}{s(s+2)} \qquad G(s) = \frac{200}{s(s+2)} = \frac{200}{s^2+2s}$

$s = j\omega \Rightarrow \frac{200}{-\cancel{j}\omega^2 + 2j\omega} = \frac{200}{-\omega(\omega - 2j)}$

modulo: $\left|\frac{200}{-\omega(\omega - 2j)}\right| = \frac{\overset{=200}{|200|}}{\underbrace{|-\omega|}_{=\omega}\underbrace{|\omega - 2j|}_{=\sqrt{\omega^2+4j^2} = \sqrt{\omega^2-4}}} = \frac{200}{\omega\sqrt{\omega^2-4}}$

$\arg\left(\frac{200}{-\omega(\omega - 2j)}\right) = \arg(200) - \arg(-\omega) - \arg(\omega - 2j) = 0° - 0° - \arctan\left(\frac{-2}{\omega}\right)$

$\arg(200) \to 0$ ↳ numeri reali; $\arg(-\omega) \to 0$ assumendo ω positiva

$\arctan(-x) = -\arctan x \Rightarrow 0 - 0 + \arctan\left(\frac{2}{\omega}\right) = \arctan\left(\frac{2}{\omega}\right)$

③ Sistema con $G(s) = \frac{K}{s(s+3)(s+5)}$ chiuso in retroazione

- Determinare i valori di K ammissibili per la stabilità

L'equazione caratteristica di un sistema a ciclo chiuso (in retroazione) è $1 + G(s) = 0$

$1 + \frac{K}{(s^2+3s)(s+5)} = (s^2+3s)(s+5) + \frac{K}{\cancel{(s^2+3s)(s+5)}}\cancel{(s^2+3s)(s+5)}$

$= K + (s^2+3s)(s+5) = s^3 + 8s^2 + 15s + K$

Criterio di Routh:

$$\begin{array}{c|ccc} 3 & 1 & 15 & 0 \\ 2 & 8 & K & 0 \\ 1 & -\frac{1}{8}(K-120) & 0 & \cdots \\ 0 & \frac{-K}{(K-120)^2} & & \end{array}$$

$x_{ik} = -\frac{1}{x_{i+1,1}}\cdot\det\left|\begin{pmatrix} x_{i+2,1} & x_{i+2,k+1} \\ x_{i+1,1} & x_{i+1,k+1} \end{pmatrix}\right|$

RICORDATI

$x_{11} = -\frac{1}{8}\cdot\det\left|\begin{pmatrix} 1 & 15 \\ 8 & K \end{pmatrix}\right| = -\frac{1}{8}\cdot(K-120)$

$x_{01} = -\frac{1}{\frac{1}{8}(K-120)}\cdot\det\left|\begin{pmatrix} 8 & K \\ \frac{1}{8}(K-120) & 0 \end{pmatrix}\right| = \frac{1}{\cancel{\frac{1}{8}}(K-120)}\left(-\frac{K}{\cancel{8}(K-120)}\right)$

$x_{01} = \frac{-K}{\underbrace{(K-120)^2}_{\text{sempre} > 0}}$

Affinché il sistema sia stabile, sulla prima colonna deve permanere il segno (positivo in questo caso)

$\Rightarrow \begin{cases} -\frac{1}{8}(K-120) > 0 \\ -K > 0 \end{cases} \Rightarrow \begin{cases} (K-120) < 0 \\ K < \end{cases} \Rightarrow \boxed{0 < K < 120}$

Per trovare i poli immaginari poniamo a 0 la riga 1:

$-\frac{1}{8}(K-120) = 0 \Rightarrow K = 120 \Rightarrow$ il sistema ha una coppia di poli puramente immaginari per $K = 120$ (è il valore che separa la stabilità ($K < 120$) dalla stabilità ($K > 120$))

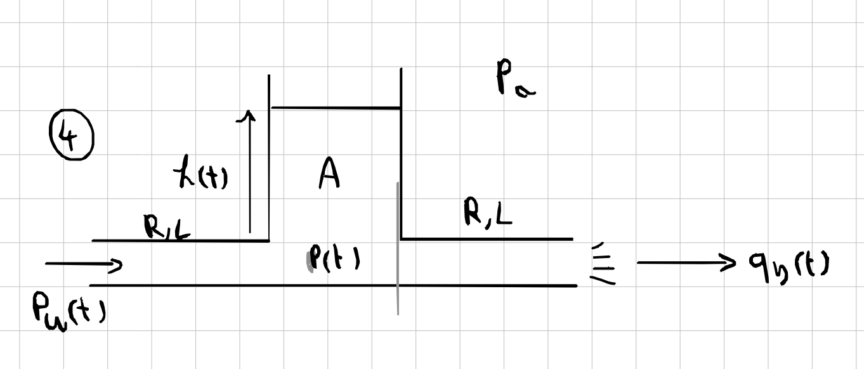
$\Rightarrow$ prendo la riga sopra: $8s^2 + K = 0$

$\Rightarrow$ sostituisco $K \Rightarrow 8s^2 + 120 = 0$

$\Rightarrow$ risolvo l'equazione $\Rightarrow s = \pm\sqrt{-15}$

poli: $p_{1,2} = \pm j\sqrt{15}$

④



Tutto lineare, $P_a = 0$

uscite: $h(t), q_y(t)$ ingresso: $P_u(t)$ Determinare A, B, C, D

Variabili: $x_1 = q_u \quad x_2 = h(t) \quad x_3 = q_y$

$P_u(t) - P(t) = R\cdot q_u + L\cdot\dot q_u \qquad P(t) = \rho\cdot g\cdot h(t)$

$\Rightarrow P_u(t) - \rho g h(t) = R\cdot q_u + L\cdot\dot q_u \Rightarrow \dot q_u = -\frac{\rho g h}{L} + \frac{P_u(t)}{L} - \frac{R}{L} q_u$

$A\cdot\dot h = q_u - q_y \Rightarrow \dot h = \frac{q_u}{A} - \frac{q_y}{A}$

$P(t) - P(a) = R\cdot q + L\cdot\dot q \Rightarrow \rho g h = R\cdot q_y + L\cdot\dot q_y \Rightarrow L\dot q_y = \rho g h - R\cdot q_y$

$\Rightarrow \dot q_y = \frac{\rho g h}{L} - \frac{R}{L}\cdot q_y$

$$\begin{cases} \dot x_1 = -\frac{\rho g}{L}\cdot x_2 - \frac{R}{L}\cdot x_1 + \frac{u}{L} \\ \dot x_2 = \frac{x_1}{A} - \frac{x_3}{A} \\ \dot x_3 = \frac{\rho g}{L} x_2 - \frac{R}{L} x_3 \end{cases} \Rightarrow A = \begin{bmatrix} -\frac{R}{L} & -\frac{\rho g}{L} & 0 \\ \frac{1}{A} & 0 & -\frac{1}{A} \\ 0 & \frac{\rho g}{L} & -\frac{R}{L} \end{bmatrix} \qquad B = \begin{bmatrix} \frac{1}{L} \\ 0 \\ 0 \end{bmatrix}$$

$y_1(t) = x_2$

$y_2(t) = x_3$

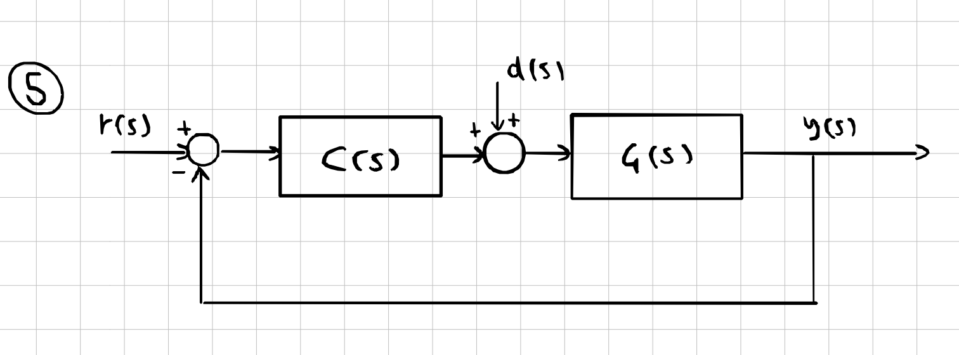
$C = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

- Calcolare lo stato di equilibrio corrispondente a $P_u = 2\,N/m^2$

$$\begin{cases} \dot x_1 = 0 \\ \dot x_2 = 0 \\ \dot x_3 = 0 \end{cases} \Rightarrow \begin{cases} -\frac{\rho g}{\cancel{L}}\cdot x_2 - \frac{R}{\cancel{L}}\cdot x_1 + \frac{2}{\cancel{L}} = 0 \\ \frac{x_1}{\cancel{A}} - \frac{x_3}{\cancel{A}} = 0 \\ \frac{\rho g}{\cancel{L}} x_2 - \frac{R}{\cancel{L}} x_3 = 0 \end{cases} \Rightarrow \begin{cases} -\rho g x_2 - \rho g x_2 + 2 = 0 \\ x_1 = x_3 \\ x_3 = \frac{\rho g}{R} x_2 \end{cases}$$

$$\Rightarrow \begin{cases} x_1 = x_3 \\ x_2 = \frac{1}{\rho g} \\ x_3 = \frac{1}{R} \end{cases} \Rightarrow \begin{cases} x_1 = \frac{1}{R} \\ x_2 = \frac{1}{\rho g} \\ x_3 = \frac{1}{R} \end{cases}$$

⑤



$C(s) = \frac{K}{s+2} \qquad G(s) = \frac{s+5}{(s+3)(s+4)}$

In serie $\Rightarrow C(s)\cdot G(s) = \frac{K(s+5)}{(s+2)(s+3)(s+4)} \equiv F(s)$

In retroazione negativa $\Rightarrow \frac{F(s)}{1 + F(s)\cdot 1} = \frac{\frac{K(s+5)}{(s+2)(s+3)(s+4)}}{1 + \frac{K(s+5)}{(s+2)(s+3)(s+4)}} =$

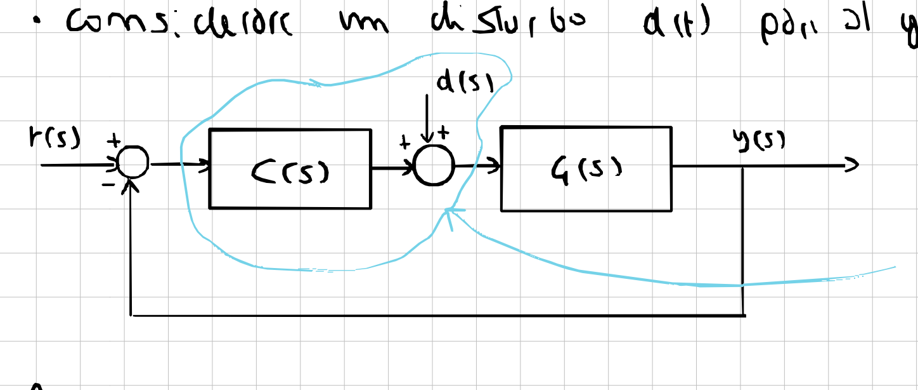
$\frac{K(s+5)}{\cancel{(s+2)(s+3)(s+4)}}\cdot\frac{\cancel{(s+2)(s+3)(s+4)}}{(s+2)(s+3)(s+4) + K(s+5)}$

$= \frac{K(s+5)}{(s+2)(s+3)(s+4) + K(s+5)}$

$Y(s) = F(s)\cdot U(s) = \frac{K(s+5)}{(s+2)(s+3)(s+4) + K(s+5)}\cdot\frac{1}{s}$

$f(\infty) = \lim_{s\to 0} s\cdot F(s) = \cancel{s}\cdot\frac{1}{\cancel{s}}\cdot\boxed{\frac{5K}{24 + 5K}}$

- considerare un disturbo d(t) pari al gradino unitario

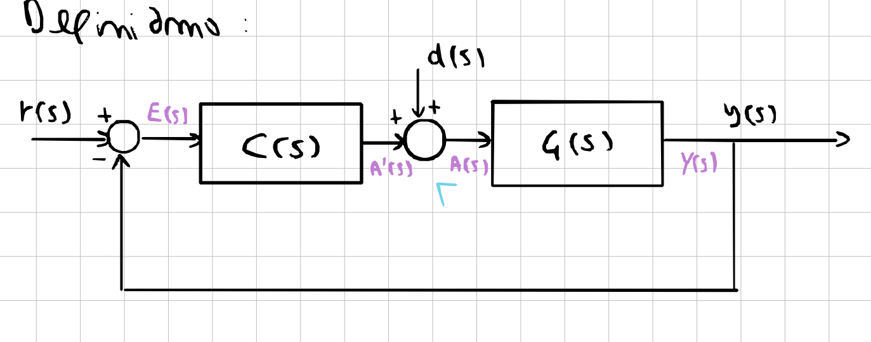


non si possono sommare una f. di trasferimento (C(s)) e un segnale (gradino unitario)

Principio di sovrapposizione degli effetti: $Y(s) = W_r(s)\cdot R(s) + W_d(s) D(s)$

$W_r(s) = \frac{C\cdot G}{1 + C\cdot G} = \frac{K(s+5)}{(s+2)(s+3)(s+4) + K(s+5)} \qquad (R(s) = 0)$

Definiamo:



$E(s) = r(s) - Y(s) = -Y(s)$

$A'(s) = C(s)\cdot E(s) = C(s)\cdot(-Y(s))$

$A(s) = A'(s) + D(s) = -C(s)Y(s) + D(s)$

$Y(s) = G(s)\cdot U(s) = G(s)(-C(s)Y(s) + D(s)) = -G(s)C(s)Y(s) + G(s)D(s)$

Essendo $Y(s) = W_r(s)\cdot R(s) + W_d(s) D(s) = W_d(s) D(s) \Rightarrow \frac{Y(s)}{D(s)} = W_d(s)$

Dunque

$Y(s) = -G(s)C(s)Y(s) + G(s)D(s) \Rightarrow Y(s)[1 + G(s)C(s)] = G(s)D(s)$

$\Rightarrow \frac{Y(s)}{D(s)} = \frac{G(s)}{1 + G(s)C(s)} = W_d(s)$

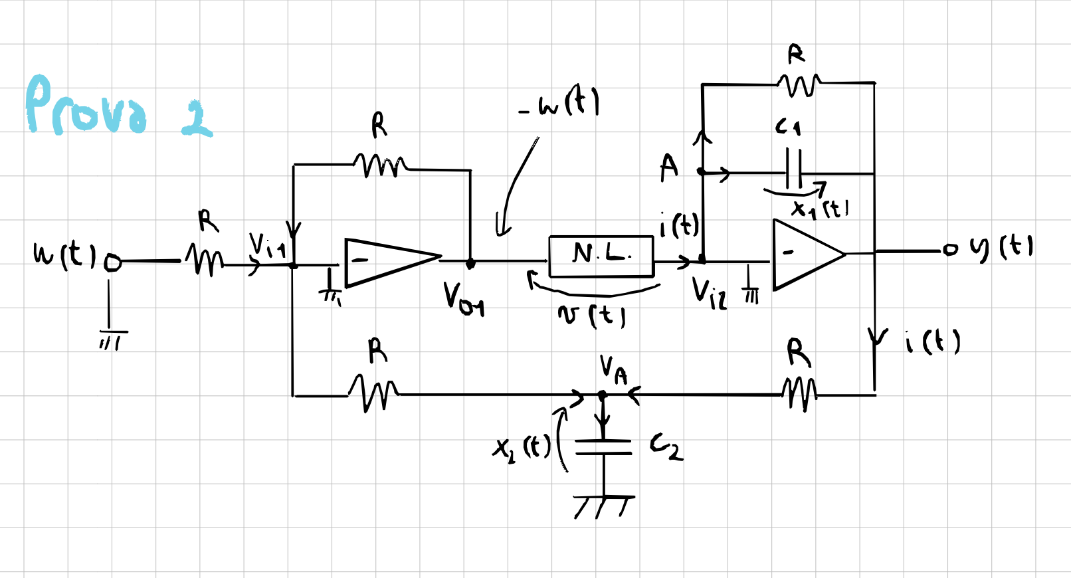
$W_d(s) = \frac{\frac{s+5}{(s+4)(s+3)}}{1 + \frac{K(s+5)}{(s+2)(s+3)(s+4)}} = \frac{\frac{s+5}{\cancel{(s+4)(s+3)}}}{\frac{(s+2)(s+3)(s+4) + K(s+5)}{(s+2)\cancel{(s+3)(s+4)}}} = \frac{(s+5)(s+2)}{(s+2)(s+3)(s+4) + K(s+5)}$

$\Rightarrow W_d(s) = \frac{10}{24 + 5K} \qquad y(\infty) = s\cdot W_d(s)\cdot D(s) = \cancel{s}\cdot W_d(s)\cdot\frac{1}{\cancel{s}} = W_d(s)$

Affinché tale risposta sia $\frac{1}{10}$ di quella trovata precedentemente,

$\frac{10}{24 + 5K} = \frac{1}{\cancel{10}\,2}\cdot\frac{\overset{1}{\cancel{5}}K}{24 + 5K} \Rightarrow \frac{20 - K}{(24 + 5K)} = 0 \Rightarrow \boxed{K = 20}$

Prova 2



V in ingresso agli opamp = 0
i " " " " = 0

Sul nodo A: $i(t) - C_1\cdot\dot{x}_1 - \frac{x_1}{R} = 0 \Rightarrow i(t) = C_1\cdot\dot{x}_1 + \frac{x_1}{R}$

$\Rightarrow \dot{x}_1 = \frac{i(t)}{C_1} - \frac{1}{RC_1}x_1$

Sappiamo che $i(t) = k_1 v + k_2 v^2$. Serve determinare v.

$v(t) = \underset{0}{V_{i2}} - V_{o1} \Rightarrow v(t) = -V_{o1} \Rightarrow V_{o1} = -v(t)$

(in ingresso all'opamp)

Scriviamo ora le LKC su V_{i1}

$\frac{V_{o1}}{R} + \frac{u}{R} + \frac{x_2}{R} = 0 \Rightarrow V_{o1} = -(u+x_2) \Rightarrow v(t) = u + x_2$

dunque $i(t) = k_1(u+x_2) + k_2(u+x_2)^2$

$\Rightarrow \dot{x}_1 = \frac{k_1}{C_1}(u+x_2) + \frac{k_2}{C_1}(u+x_2)^2 - \frac{1}{RC_1}\cdot x_1$

Applico ora LKC su V_A

$C_2\cdot\dot{x}_2 + \frac{x_2}{R} + \frac{x_2-y}{R} = 0$ y è x_1 $\Rightarrow \dot{x}_2 = -\frac{x_2}{C_2R} - \frac{(x_2-x_1)}{C_2R}$

$\Rightarrow \dot{x}_2 = -\frac{x_2}{C_2R} - \frac{x_2}{C_2R} + \frac{x_1}{C_2R} = -\frac{2x_2}{C_2R} + \frac{x_1}{C_2R}$

- $R=1$, $K_1=4$, $K_2=2$. Calcolare gli stati di equilibrio corrispondenti ad ingresso nullo ($u=0$)

$$\begin{cases} \overset{f_1}{\frac{4}{C_1}x_2 + \frac{2}{C_1}x_2^2 - \frac{1}{C_1}x_1 = 0} \\ \underset{f_2}{-\frac{2x_2}{C_2} + \frac{x_1}{C_2} = 0} \end{cases} \qquad \begin{cases} 2x_2^2 + 2x_2 = 0 \\ x_1 = 2x_2 \end{cases} \qquad x = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad x = \begin{Bmatrix} -2 \\ -1 \end{Bmatrix}$$

i segni risultano opposti ma non tornano, forse errore del prof

- Stabilità degli stati di equilibrio (possiamo semplificare i C_1 e C_2 ai denominatori)

$f_1 = 4x_2 - 2x_2^2 - x_1$ $\qquad f_2 = -2x_2 + x_1$

$\frac{\partial f_1}{\partial x_1} = -1 \quad \frac{\partial f_1}{\partial x_2} = 4-4x_2 \quad \frac{\partial f_2}{\partial x_1} = 1 \quad \frac{\partial f_2}{\partial x_2} = -2$

$$A = \begin{bmatrix} -1 & 4-4x_2 \\ 1 & -2 \end{bmatrix} \qquad (\lambda I - A) = \begin{bmatrix} \lambda+1 & 4-4x_2 \\ 1 & \lambda+2 \end{bmatrix} = \lambda^2 + 2\lambda + \lambda + 2 - 4 + 4x_2$$

$\Rightarrow \lambda^2 + 3\lambda - 2 + 4x_2$

in $\begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \Rightarrow \lambda^2 + 3\lambda - 2 = 0 \Rightarrow \frac{-3 \pm \sqrt{4+4\cdot 2}}{2} = \frac{-3\pm\sqrt{17}}{2} \Rightarrow$ positivi

$\Rightarrow$ instabile

in $\begin{Bmatrix} 2 \\ 1 \end{Bmatrix} \Rightarrow \lambda^2 + 3\lambda - 2 + 4 = \lambda^2 + 3\lambda + 2 = 0 \Rightarrow \frac{-3\pm\sqrt{9-8}}{2}$ $\nearrow -2$ $\searrow -1$

quello che ha trovato il prof

$\Rightarrow$ asintoticamente stabile

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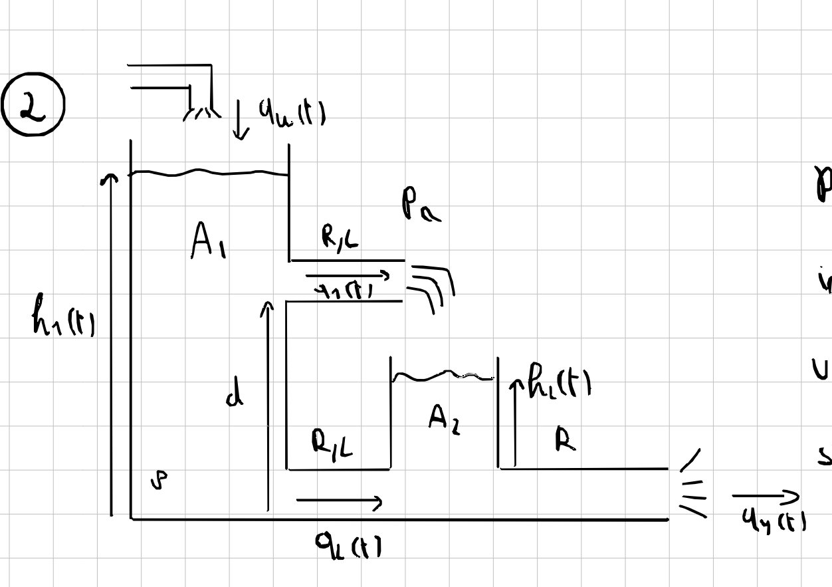


$P_a = 0$

ingressi: $q_u(t)$, d (distanza fisso)

uscite: $h_1(t)$, $h_2(t)$, $q_y(t)$

stati: $h_1(t)$, $h_2(t)$, $q_1(t)$, $q_2(t)$

$P_2(t) - P_a = R\cdot q_y \Rightarrow \rho\cdot g\cdot h_2 = R\cdot q_y$ $\qquad q_y = \frac{\rho\cdot g\cdot h_2}{R}$

$A_1\cdot\dot{h}_1 = q_u(t) - q_1(t) - q_2(t)$

$A_2\cdot\dot{h}_2 = q_2(t) + q_1(t) - \frac{\rho\cdot g\cdot h_2}{R}$

PRESSIONE IN CORRISPONDENZA DI q_1

$P_1(t) - P_a = R\cdot q_1 + L\cdot\dot{q}_1 \qquad P(t) = \rho\cdot g\cdot h_1(t) \Rightarrow \rho\cdot g\,\boxed{(h_1(t) - d)} = R\cdot q_1 + L\cdot\dot{q}_1$

$P_1(t) - P_2(t) = R\cdot q_2 + L\cdot\dot{q}_2 \Rightarrow \rho g h_1 - \rho g h_2 = R\cdot q_2 + L\cdot\dot{q}_2$

$$\Rightarrow \begin{cases} \dot{h}_1 = \frac{q_u(t)}{A} - \frac{q_1(t)}{A} - \frac{q_2(t)}{A} \\ \dot{h}_2 = \frac{q_2(t)}{A} + \frac{q_1(t)}{A} - \frac{\rho\cdot g\cdot h_2}{RA} \\ \dot{q}_1 = \frac{\rho\cdot g\cdot h_1}{L} - \frac{\rho g}{L}u_2 - \frac{R}{L}q_1 \\ \dot{q}_2 = \frac{\rho g h_1}{L} - \frac{\rho g h_2}{L} - \frac{R}{L}q_2 \end{cases} \Rightarrow \begin{cases} \dot{x}_1 = \frac{u_1}{A} - \frac{x_2}{A} - \frac{x_3}{A} \\ \dot{x}_2 = \frac{\rho g}{L}\cdot x_1 - \frac{R}{L}\cdot x_2 - \frac{\rho g}{L}u_2 \\ \dot{x}_3 = \frac{\rho\cdot g}{L}\cdot x_1 - \frac{\rho\cdot g}{L}x_2 - \frac{R}{L}x_3 \\ \dot{x}_4 = \frac{x_3}{A} + \frac{x_2}{A} - \frac{\rho\cdot g}{A}\cdot x_4 \end{cases}$$

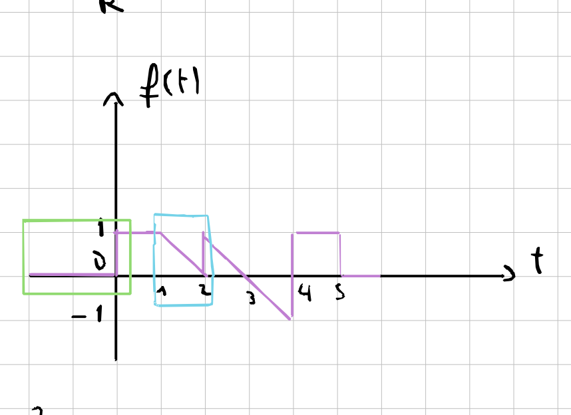
$x_1 = h_1$, $x_2 = q_1$, $x_3 = q_2$, $x_4 = h_2$

$$\Rightarrow A = \begin{bmatrix} 0 & -\frac{1}{A_1} & -\frac{1}{A_1} & 0 \\ \frac{\rho g}{L} & -\frac{R}{L} & 0 & 0 \\ \frac{\rho g}{L} & 0 & -\frac{R}{L} & -\frac{\rho g}{L} \\ 0 & \frac{1}{A_2} & \frac{1}{A_2} & -\frac{\rho g}{RA_2} \end{bmatrix} \qquad B = \begin{bmatrix} \frac{1}{A} & 0 \\ 0 & -\frac{\rho g}{L} \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

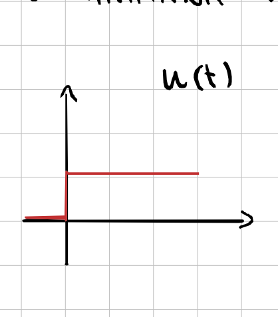
$y_1 = h_1$
$y_2 = h_2$
$y_3 = q_y = \frac{\rho g}{R}\cdot h_2$

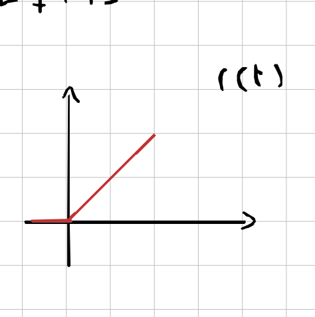
$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & \frac{\rho g}{R} \end{bmatrix} \qquad D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

③



determinare $\mathcal{L}[f(t)]$





$f(t) = ?$

- $t=0$ da 0 a 1 $\Rightarrow A = 1$ $\quad m = 0$ (prima era piatto, dopo è piatto) $\Rightarrow u(t)$
- $t=1$ $\quad m = \frac{y_2-y_1}{x_2-x_1} = \frac{1-0}{1-2} = -1 \Rightarrow -r(t-1)$
- $t=2$ salto $\Rightarrow u(t-2)$ $\quad m_{precedente} = -1$, $m_{successiva}$ (tra 2 e 4) $= \frac{1-(-1)}{2-4} = -1$

  $\Delta m = 0 \Rightarrow$ non va aggiunta una rampa
- $t=4$ salto $\Rightarrow 2\cdot u(t-4)$, dopo il salto, la pendenza è 0 ($m_{successiva} = 0$)

  $\Rightarrow \Delta m = 0 - (-1) = 1 \Rightarrow$ va aggiunta una rampa $\Rightarrow 2u(t-4) + r(t-4)$
- $t=5$ salto negativo $\quad m_{prec} = 0$, $m_{succ} = 0 \Rightarrow -u(t-5)$ niente rampa

$\Rightarrow f(t) = u(t) - r(t-1) + u(t-2) + 2u(t-4) + r(t-4) - u(t-5)$

$r = t\cdot h(t) \qquad u(t) = h(t)$

$\Rightarrow f(t) = h(t) - (t-1)h(t-1) + h(t-2) + 2h(t-4) + (t-4)h(t-4) - h(t-5)$

$\mathcal{L}[f(t)] = \frac{1}{s} - \frac{1}{s^2}\cdot e^{-s} + \frac{1}{s}\cdot e^{-2s} + \frac{2}{s}\cdot e^{-4s} + \frac{1}{s^2}\cdot e^{-4s} - \frac{1}{s}\cdot e^{-5s}$

ricordando che $\mathcal{L}[f(t-T_0)] = F(s)\cdot e^{-T_0 s}$

④ $\quad \dot{x}(t) = Ax(t) + Bu(t) \qquad A = \begin{bmatrix} -2 & 0 \\ 4 & -5 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad C = [0 \;\; 1]$

$y(t) = Cx(t)$

Sia u nullo, calcolare $x(3)$ sapendo che $x(0) = \begin{bmatrix} \frac{3}{4} \\ 1 \end{bmatrix}$

$$\begin{cases} \dot{x}_1 = -2x_1 \\ \dot{x}_2 = 4x_1 - 5x_2 \\ y(t) = x_2 \end{cases} \longleftarrow \text{ingressi nulli}$$

$x(t) = e^{At}x_0$

$e^{At} = \mathcal{L}^{-1}[(sI-A)^{-1}] \qquad sI - A = \begin{bmatrix} s+2 & 0 \\ -4 & s+5 \end{bmatrix}$

RICORDATI CHE È $\begin{bmatrix} s-\dots & 0-\dots \\ 0-\dots & s-\dots \end{bmatrix}$

si invertono anche i segni dove non c'è s

$\det(sI-A) = (s+2)(s+5)$

$\operatorname{agg}(sI-A)$

$C_{ij} = (-1)^{i+j}\cdot\det(M_{ij})$

$C_{11} = 1\cdot(s+5) = s+5 \quad C_{12} = -1(-4) = +4 \quad C_{21} = 0 \quad C_{22} = 1\cdot s+2 = s+2$

$\Rightarrow C = \begin{bmatrix} s+5 & -4 \\ 0 & s+2 \end{bmatrix} \qquad C^T = \begin{bmatrix} s+5 & 0 \\ +4 & s+2 \end{bmatrix}$

$$(sI-A)^{-1} = \frac{\operatorname{agg}}{\det} = \begin{bmatrix} \frac{\cancel{(s+5)}}{(s+2)\cancel{(s+5)}} & 0 \\ \frac{-4}{(s+2)(s+5)} & \frac{\cancel{(s+2)}}{\cancel{(s+2)}(s+5)} \end{bmatrix} = \begin{bmatrix} \frac{1}{s+2} & 0 \\ \frac{+4}{(s+2)(s+5)} & \frac{1}{s+5} \end{bmatrix}$$

$\mathcal{L}^{-1}\left[\frac{1}{s+2}\right] = e^{-2t}$

$\mathcal{L}^{-1}\left[\frac{1}{s+5}\right] = e^{-5t}$

$$\frac{4}{(s+2)(s+5)} = \frac{4}{(s-(-2))(s-(-5))} = \frac{A}{(s-(-2))} + \frac{B}{(s-(-5))} = \frac{A\cdot(s-(-5)) + B\cdot(s-(-2))}{(s-(-5))(s-(-2))}$$

$\lambda_1 = -2$, $\lambda_2 = -5$

$A(s+5) + B(s+2) = 4 \Rightarrow As + 5A + Bs + 2B = 4 \Rightarrow (A+B)\cdot s + (5A+2B) = 0\cdot s + 4$

$$\begin{cases} A+B = 0 \\ 5A+2B = 4 \end{cases} \Rightarrow \begin{cases} A = -B \\ -5B+2B = 4 \end{cases} \Rightarrow \begin{cases} A = \frac{4}{3} \\ B = -\frac{4}{3} \end{cases}$$

$\Rightarrow \frac{4/3}{(s+2)} - \frac{4/3}{(s+5)} \Rightarrow \frac{4}{3}\cdot\left(\frac{1}{s+2} - \frac{1}{s+5}\right)$

$\mathcal{L}^{-1}\left[\frac{4}{3}\cdot\left(\frac{1}{s+2} - \frac{1}{s+5}\right)\right] = \frac{4}{3}\cdot\mathcal{L}^{-1}\left[\left(\frac{1}{s+2} - \frac{1}{s+5}\right)\right] = \frac{4}{3}e^{-2t} - \frac{4}{3}e^{-5t}$

$$\Rightarrow x(t) = \begin{bmatrix} e^{-2t} & 0 \\ \frac{4}{3}e^{-2t} - \frac{4}{3}e^{-5t} & e^{-5t} \end{bmatrix}\cdot x_0 = \begin{bmatrix} e^{-2t} & 0 \\ \frac{4}{3}e^{-2t} - \frac{4}{3}e^{-5t} & e^{-5t} \end{bmatrix}\cdot\begin{Bmatrix} \frac{3}{4} \\ 1 \end{Bmatrix}$$

$$\begin{Bmatrix} e^{-2t}\left(\frac{3}{4}\right) + 0\cdot(1) \\ \cancel{\frac{4}{3}}e^{-2t} - \cancel{\frac{4}{3}e^{-5t}}\cdot\left(\cancel{\frac{3}{4}}\right) + \cancel{e^{-5t}}\cdot(1) \end{Bmatrix}\Bigg|_{t=3} \Rightarrow \begin{Bmatrix} \frac{3}{4}e^{-6} \\ e^{-6} \end{Bmatrix}$$

- Determinare $g(t)$

$$\begin{cases} \dot{x}_1 = -2x_1 + u \\ \dot{x}_2 = 4x_1 - 5x_2 + 2u \\ y(t) = x_2 \end{cases}$$

$g(t) = \mathcal{L}^{-1}[G(s)] \qquad G(s) = C\cdot(sI-A)^{-1}B + \overset{0}{D}$

$$= [0 \;\; 1]\cdot\begin{bmatrix} \frac{1}{s+2} & 0 \\ \frac{+4}{(s+2)(s+5)} & \frac{1}{s+5} \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = [0 \;\; 1]\begin{bmatrix} \frac{1}{s+2} & 0 \\ \frac{4}{(s+2)(s+5)} + \frac{2}{s+5} \end{bmatrix}$$

$$= [0 \;\; 1]\begin{bmatrix} \frac{1}{s+2} \\ \frac{2s+8}{(s+2)(s+5)} \end{bmatrix} = \frac{2s+8}{(s+2)(s+5)}$$

$\frac{A}{s+2} + \frac{B}{s+5} \Rightarrow A(s+5) + B(s+2) = 2s+8 \Rightarrow sA + 5A + sB + 2B = 2s+8$

$$\Rightarrow (A+B)s + (5A+2B) = 2s+8 \Rightarrow \begin{cases} A+B = 2 \\ 5A+2B = 8 \end{cases} \Rightarrow \begin{cases} A = 2-B \\ 10-5B+2B = 8 \end{cases}$$

$$\Rightarrow \begin{cases} A = 2-B \\ -3B = -2 \end{cases} \Rightarrow \begin{cases} A = 2-\frac{2}{3} \\ B = \frac{2}{3} \end{cases} \Rightarrow \begin{cases} A = \frac{4}{3} \\ B = \frac{2}{3} \end{cases}$$

$\Rightarrow g(t) = \frac{4}{3}e^{-2t} - \frac{2}{3}e^{-5t}$

- modulo e argomento risposta armonica:

$$G(s) = \frac{2s+8}{(s+2)(s+5)} = \frac{\overset{4}{\cancel{8}}\left(\frac{1}{4}s+1\right)}{\cancel{2}\left(\frac{s}{2}+1\right)5\left(\frac{s}{5}+1\right)} = \frac{\frac{4}{5}\cdot\left(\frac{1}{4}s+1\right)}{\left(\frac{s}{2}+1\right)\left(\frac{s}{5}+1\right)}$$

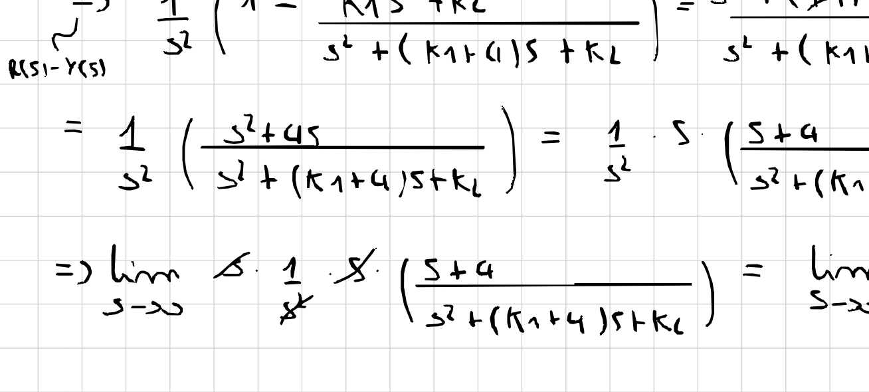
$$G(j\omega) = \frac{\frac{4}{5}\left(\frac{1}{4}j\omega+1\right)}{\left(\frac{1}{2}j\omega+1\right)\left(\frac{1}{5}j\omega+1\right)} = |G(j\omega)| = \frac{\left|\frac{4}{5}\right|\left|\frac{1}{4}j\omega+1\right|}{\left|\frac{1}{2}j\omega+1\right|\left|\frac{1}{5}j\omega+1\right|}$$

$\frac{1}{25}\omega^2+1$

$\frac{1}{4}\omega^2+1$

$$= \frac{\frac{4}{5}\cdot\sqrt{\frac{1}{16}\omega^2+1}}{\sqrt{\frac{1}{4}\omega^2+1}\sqrt{\frac{1}{25}\omega^2+1}}$$

$$\arg(G(j\omega)) = \arg\left(\frac{\frac{4}{5}\left(\frac{1}{4}j\omega+1\right)}{\left(\frac{1}{2}j\omega+1\right)\left(\frac{1}{5}j\omega+1\right)}\right)$$



$R(s) = K_1 + \frac{K_2}{s} \qquad G(s) = \frac{1}{s+4}$

$R(s) = \frac{K_1 s + K_2}{s} \qquad R(s)\cdot G(s) = \frac{K_1 s + K_2}{s(s+4)} \equiv F(s)$

$$\text{Retroazione negativa} \Rightarrow \frac{F(s)}{1+F(s)} = \frac{\frac{K_1 s+K_2}{s(s+4)}}{1 + \frac{K_1 s+K_2}{s(s+4)}} = \frac{\frac{K_1 s+K_2}{s(s+4)}}{\frac{s(s+4)+K_1 s+K_2}{s(s+4)}}$$

$$= \frac{K_1 s + K_2}{s(s+4)+K_1 s+K_2} = \frac{K_1 s+K_2}{s^2+(K_1+4)s+K_2}$$

$e(t) = r(t) - y(t)$

$\lim_{t\to\infty} e(t) = \lim_{s\to 0} s\cdot E(s) = \lim_{s\to 0} s\left(R(s) - Y(s)\right)$

$Y(s) = W\cdot r(s) = \frac{1}{s^2}\frac{K_1 s+K_2}{s^2+(K_1+4)s+K_2}$

$$\underset{R(s)-Y(s)}{\Rightarrow} \frac{1}{s^2}\left(1 - \frac{K_1 s+K_2}{s^2+(K_1+4)s+K_2}\right) = \frac{s^2 + (\cancel{K_1}+4)s + \cancel{K_2} - \cancel{K_1 s} - \cancel{K_2}}{s^2+(K_1+4)s+K_2}$$

$$= \frac{1}{s^2}\left(\frac{s^2+4s}{s^2+(K_1+4)s+K_2}\right) = \frac{1}{s^2}\cdot s\cdot\left(\frac{s+4}{s^2+(K_1+4)s+K_2}\right)$$

$$\Rightarrow \lim_{s\to\infty} \cancel{s}\cdot\frac{1}{\cancel{s^2}}\cdot\cancel{s}\cdot\left(\frac{s+4}{s^2+(K_1+4)s+K_2}\right) = \lim_{s\to\infty}\left(\frac{s+4}{s^2+(K_1+4)s+K_2}\right) = \frac{4}{K_2}$$

$\frac{4}{K_2} = 0.01 \Rightarrow \boxed{K_2 = 400}$ da capire se è giusto il procedimento

- La funzione di trasferimento è $\frac{K_1 s+K_2}{s^2+(K_1+4)s+K_2}$.

Un sistema del secondo ordine con poli complessi coniugati ha il denominatore nella forma $s^2 + 2\delta\omega_n s + \omega_n^2$

$$\text{dunque} \begin{cases} K_1+4 = 2\delta\omega_n \\ \omega_n^2 = K_2 \end{cases} \Rightarrow \begin{cases} \text{"} \\ \omega_n = \sqrt{400} \end{cases} \Rightarrow \begin{cases} K_1+4 = 2\,\delta\cdot 20 \\ \omega_n = 20 \end{cases}$$

Il problema chiede di far sì che δ sia $\frac{1}{2}$

$\Rightarrow K_1 + 4 = 20 \Rightarrow \boxed{K_1 = 16}$

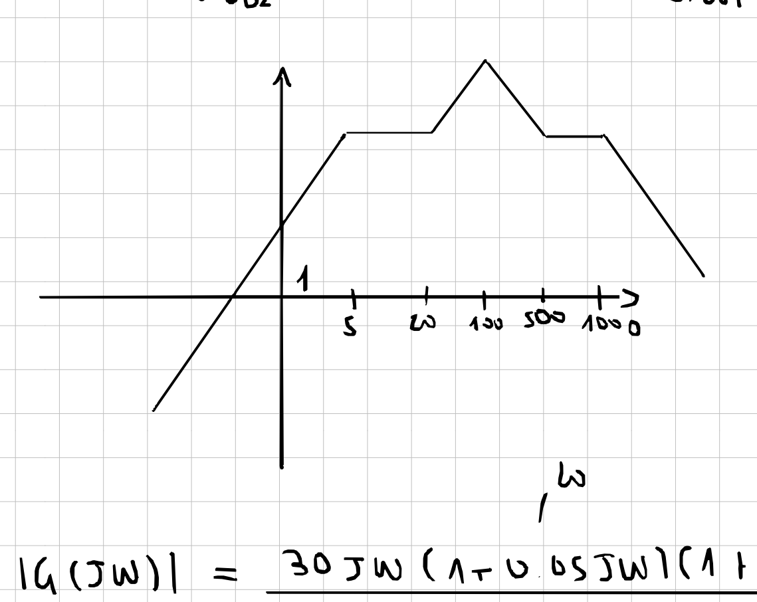
⑥ $\quad G(s) = \frac{30s(1+0.05s)(1+0.002s)}{(1+0.2s)(1+0.01s)^2(1+0.001s)}$

$G(j\omega) = \frac{30j\omega(1+0.05j\omega)(1+0.002j\omega)}{(1+0.2j\omega)(1+0.01j\omega)^2(1+0.001j\omega)}$

$\omega_{z0} = 0 \qquad \omega_{p1} = \frac{1}{0.2} = -5$

$\omega_{z1} = \frac{1}{0.05} = -20 \qquad \omega_{p2} = \omega_{p3} = \frac{1}{0.01} = -100$

$\omega_{z2} = \frac{1}{0.002} = -500 \qquad = \frac{1}{0.001} = 1000$



$$|G(j\omega)| = \frac{30j\omega(1+0.\overset{20}{05}j\omega)(1+0.\overset{500}{002}j\omega)}{(1+0.\underset{5}{2}j\omega)(1+0.\underset{100}{01}j\omega)^2(1+0.\underset{1000}{001}j\omega)}$$

$|G(j5)| = 30\cdot 5 = 150$

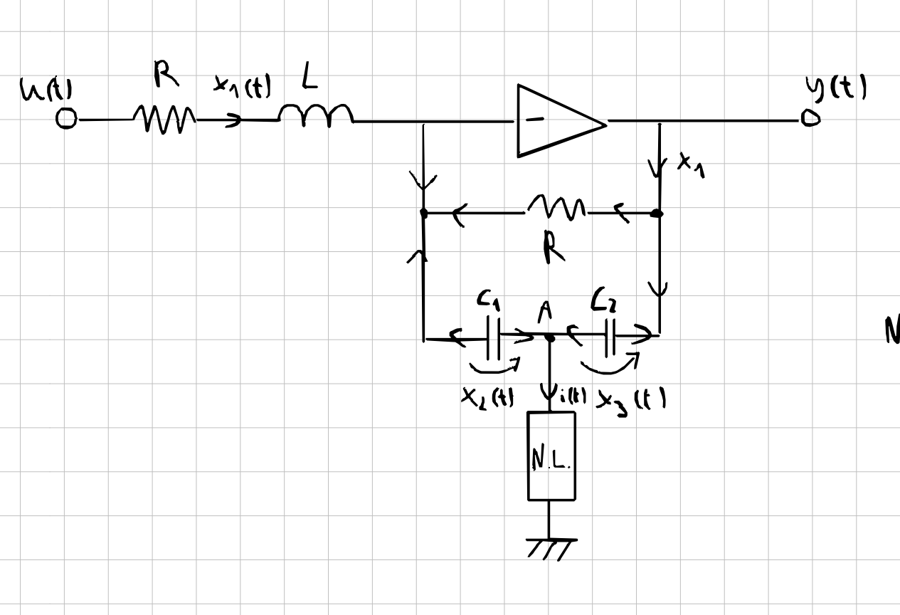
$|G(j20)| = \frac{30\cdot\overset{5}{\cancel{20}}}{\cancel{4}} = 150$

$|G(j100)| = \frac{30\cdot 100\cdot(0.05\cdot 100)}{0.2\cdot 100} = \frac{30\cdot\overset{5}{\cancel{100}}\cdot 5}{\cancel{20}\cdot 1} = 25\cdot 30 = 750$

$|G(j500)| = \frac{30\cdot 5\cancel{00}\cdot\left(\frac{5}{100}\cdot 5\cancel{00}\right)}{\frac{2}{10}\cdot\cancel{500}\cdot\left(\frac{1}{100}\cdot 500\right)^2} = \frac{150\cdot\cancel{25}}{\cancel{25}} = 150$

$|G(j1000)| = \frac{30\cdot 100\cdot(0.05\cdot\cancel{1000})\cdot(0.002\cdot\cancel{1000})}{(0.2\cdot\cancel{1000})(0.01\cdot\cancel{1000})^2} = \frac{30\cdot 5\cdot\cancel{0.002}}{\frac{2}{10}\cdot\frac{1}{100}} = 150$

Prova 3



N.L. : $i(t) = K_1 v(t) + K_2 v^3(t)$

$$u - Rx_1 - L\dot{x}_1 = 0 \quad \Rightarrow \quad \dot{x}_1 = \frac{u}{L} - \frac{R}{L}x_1$$

$$C_1\dot{x}_2 + x_1 + \frac{(x_3+x_2)}{R} = 0$$

$$-C_1\dot{x}_2 + C_2\dot{x}_3 - i(t) = 0 \Rightarrow -C_1\dot{x}_2 + C_2\dot{x}_3 - K_1 v(t) - K_2 v^3(t) = 0$$

$$v(t) = x_2 \Rightarrow C_1\dot{x}_2 + C_2\dot{x}_3 - K_1x_2 - K_2x_2^3 = 0$$

$$\Rightarrow \quad \dot{x}_1 = \frac{u}{L} - \frac{R}{L}x_1$$

$$\dot{x}_2 = -\frac{x_1}{C_1} - \frac{(x_3+x_2)}{RC_1}$$

$$-\cancel{C_1}\left(-\frac{x_1}{\cancel{C_1}} - \frac{(x_3+x_2)}{R\cancel{C_1}}\right) + C_2\dot{x}_3 - K_1x_2 - K_2x_2^3 \Rightarrow$$

$$C_2\dot{x}_3 = -x_1 - \frac{(x_3+x_2)}{R} + K_1x_2 + K_2x_2^3$$

$$\dot{x}_3 = -\frac{x_1}{C_2} - \frac{(x_3+x_2)}{RC_2} + \frac{K_1x_2}{C_2} + \frac{K_2x_2^3}{C_2}$$

$y(t) = x_2(t) + x_3(t)$

- $\bar{u} = 0 \Rightarrow \begin{cases} \dot{x}_1 = 0 \Rightarrow -\frac{R}{L}x_1 = 0 \Rightarrow x_1 = 0 \\ \dot{x}_2 = 0 \Rightarrow -\frac{(x_3+x_2)}{RC_1} = 0 \Rightarrow x_3 = -x_2 \\ \dot{x}_3 = 0 \Rightarrow \frac{K_1x_2}{C_2} + \frac{K_2x_2^3}{C_2} = 0 \Rightarrow x_2(K_1 + K_2x_2^2) = 0 \end{cases}$

$x_2 = 0$; $x_2^2 = -\frac{K_1}{K_2}$

$$\bar{x} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}$$

(+ 2 stati di equilibrio che però non sono richiesti dal problema)
(questo è quello "ovvio")

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \frac{\partial f_1}{\partial x_3} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \frac{\partial f_2}{\partial x_3} \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} & \frac{\partial f_3}{\partial x_3} \end{bmatrix}$$

$$\frac{\partial f_1}{\partial x_1} = -\frac{R}{L} \qquad \frac{\partial f_1}{\partial x_2} = 0 \qquad \frac{\partial f_1}{\partial x_3} = 0$$

$$\frac{\partial f_2}{\partial x_1} = -\frac{1}{C_1} \qquad \frac{\partial f_2}{\partial x_2} = -\frac{1}{RC_1} \qquad \frac{\partial f_3}{\partial x_3} = -\frac{1}{RC_1}$$

$$\frac{\partial f_3}{\partial x_1} = -\frac{1}{C_2} \qquad \frac{\partial f_3}{\partial x_2} = \frac{1}{RC_2} + \frac{K_1}{C_2} + \frac{3x_2^2}{C_2} \qquad \frac{\partial f_3}{\partial x_3} = \frac{-1}{RC_2}$$

$$\Rightarrow A = \begin{bmatrix} -\frac{R}{L} & 0 & 0 \\ -\frac{1}{C_1} & -\frac{1}{RC_1} & -\frac{1}{RC_1} \\ -\frac{1}{C_2} & -\frac{1}{RC_2} + \frac{K_1}{C_2} + \frac{3x_2^2}{C_2} & -\frac{1}{RC_2} \end{bmatrix} \Rightarrow \text{in } \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix} \Rightarrow \begin{bmatrix} -\frac{R}{L} & 0 & 0 \\ -\frac{1}{C_1} & -\frac{1}{RC_1} & -\frac{1}{RC_1} \\ -\frac{1}{C_2} & -\frac{1}{RC_2} + \frac{K_1}{C_2} & -\frac{1}{RC_2} \end{bmatrix}$$

$$B = \begin{bmatrix} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \\ \frac{\partial f_3}{\partial u} \end{bmatrix} \Rightarrow B = \begin{bmatrix} \frac{1}{L} \\ 0 \\ 0 \end{bmatrix}$$

$y(t) = x_2(t) + x_3(t) \Rightarrow C = [\,0 \ \ 1 \ \ 1\,] \qquad D = 0$

② $G(s) = \frac{3}{2s+1}$ $\qquad u(t) = 2\sin\left(\frac{\sqrt{3}}{2}t\right) + \frac{2}{\sqrt{3}}\cos\left(\frac{1}{2\sqrt{3}}t\right)$ $\qquad y(\infty) = ?$

$\lim_{t\to\infty} y(t) = \lim_{s\to 0} s \cdot G(s) \cdot U(s)$ (teorema del valore finale)

$$U(s) = \frac{2 \cdot \frac{\sqrt{3}}{2}}{s^2 + \frac{3}{4}} + \frac{2}{\sqrt{3}} \cdot \frac{s}{s^2 + \frac{1}{12}}$$

$$s \cdot U(s) \cdot G(s) = \left(\frac{\frac{3}{2s+1} \cdot \cancel{2} \cdot \frac{\sqrt{3}}{\cancel{2}}}{s^2 + \frac{3}{4}} + \frac{2}{\sqrt{3}} \cdot \frac{s}{\frac{12s^2+1}{12}} \cdot \frac{\overset{\sqrt{3}}{\cancel{3}}}{2s+1} \right) \cdot s$$

$$= \frac{\left(\frac{3\sqrt{3}s}{2s+1}\right)}{\left(\frac{4s^2+3}{4}\right)} + \left(\frac{\cancel{2}\sqrt{3}s^2}{(12s^2+1)(2s+1)}\right)_{\cancel{12}\,6} = \frac{12\sqrt{3}s}{(4s^2+3)(2s+1)} + \frac{6\sqrt{3}s^2}{(12s^2+1)(2s+1)}$$

$$\Rightarrow \frac{As+B}{(4s^2+3)} + \frac{C}{(2s+1)} = 12\sqrt{3}s \Rightarrow As^2 + As + 2Bs + B + 4Cs^2 + 3C = 12\sqrt{3}s$$

$$\begin{cases} A + 4C = 0 \\ A + 2B = 12\sqrt{3} \\ B + 3C = 0 \end{cases} \Rightarrow \begin{cases} A = -4C \\ -4C = 12\sqrt{3} - 2B \\ '' \end{cases} \Rightarrow \begin{cases} '' \\ C = \frac{B}{2} - 3\sqrt{3} \\ B + \frac{3}{2}B - 9\sqrt{3} = 0 \end{cases}$$

$$= \begin{cases} ' \\ '' \\ \frac{5}{2}B = 9\sqrt{3} \end{cases} \Rightarrow \begin{cases} C = \frac{9}{5}\sqrt{3} - 3\sqrt{3} \\ B = \frac{18}{5}\sqrt{3} \end{cases} \Rightarrow \begin{cases} A = \frac{24}{5}\sqrt{3} \\ C = -\frac{6}{5}\sqrt{3} \\ B = \frac{18}{5}\sqrt{3} \end{cases}$$

$$\Rightarrow \frac{As+B}{(12s^2+1)} + \frac{C}{2s+1} = 6\sqrt{3}s^2 \Rightarrow 2As^2 + As + 2Bs + B + 12Cs^2 + C = 6\sqrt{3}s^2$$

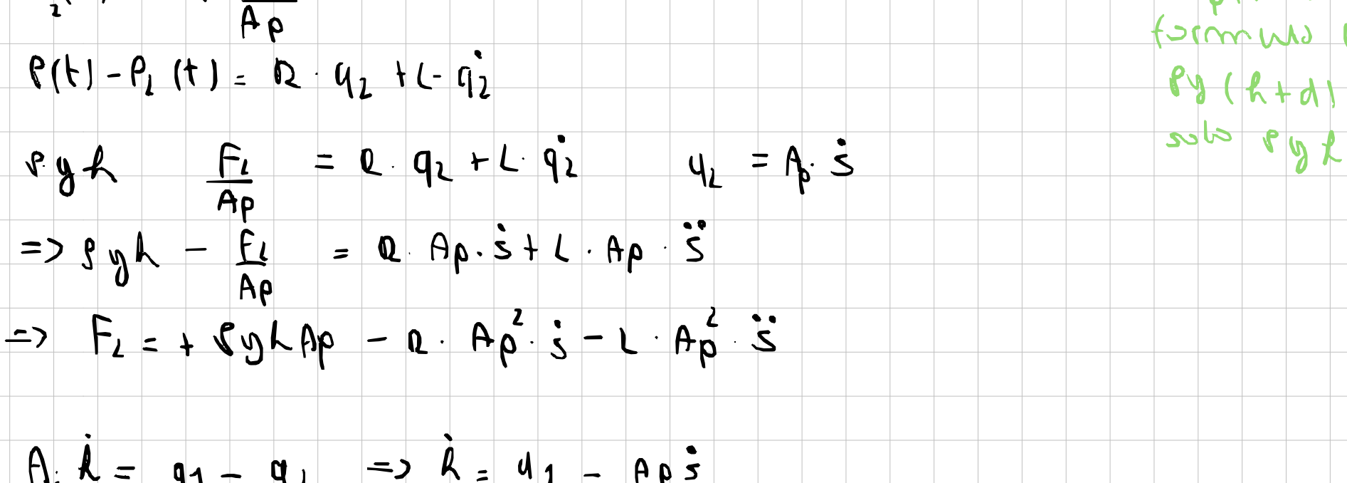
$$\begin{cases} 2A + 12C = 6\sqrt{3} \\ A + 2B = 0 \\ B + C = 0 \end{cases} \Rightarrow \begin{cases} 4C + 12C = 6\sqrt{3} \\ A = 2C \\ B = -C \end{cases} \Rightarrow \begin{cases} C = \frac{3}{8}\sqrt{3} \\ A = \frac{3}{4}\sqrt{3} \\ B = -\frac{3}{8}\sqrt{3} \end{cases}$$

$$\Rightarrow Y(s) = \frac{\left(\frac{24}{5}\sqrt{3}\right)s}{4s^2+3} + \frac{\left(\frac{18}{5}\sqrt{3}\right)}{4s^2+3} - \frac{\left(\frac{6}{5}\sqrt{3}\right)}{2s+1} + \frac{\left(\frac{3}{4}\sqrt{3}\right)s}{12s^2+1} - \frac{\left(\frac{3}{8}\sqrt{3}\right)}{12s^2+1} + \frac{\left(\frac{3}{8}\sqrt{3}\right)}{2s+1}$$

$$\Rightarrow y(t) = \frac{\overset{6}{\cancel{24}}}{\cancel{4}\cdot 5}\sqrt{3}\cos\left(\frac{\sqrt{3}}{2}t\right) + \frac{18}{5\cdot 4}\sqrt{3}\,\text{sen}\left(\frac{\sqrt{3}}{2}t\right) - \frac{\overset{3}{\cancel{6}}}{5\cdot\cancel{2}}\sqrt{3}\,\text{sen}\left(\frac{t}{\sqrt{2}}\right) + \frac{\overset{1}{\cancel{3}}}{4\cdot\cancel{12}4}\sqrt{3}\cos\left(\frac{1}{\sqrt{12}}t\right)$$

$$-\frac{3}{8\cdot 12}\sqrt{3}\,\text{sen}\left(\frac{1}{2\sqrt{3}}t\right) + \frac{3}{8\cdot 2}\sqrt{3}\left(\frac{t}{\sqrt{2}}\right)$$

Non viene, sarebbe da risolvere con la frequenza (e probabilmente i calcoli sono sbagliati)



$x_1 = q_1(t) \quad x_2 = h(t) \quad x_3 = s(t) \quad x_4 = \dot{s}(t) \quad u_1 = F(t) \quad u_2 = d \quad y_1 = s(t) \quad y_2 = h(t)$

$$P_1(t) = \frac{F(t)}{Ap} \qquad P(t) = P_a + \rho\cdot g\cdot(d + h(t)) \Rightarrow P(t) = \rho\cdot g\cdot(d + h(t))$$

$$P_1(t) - P(t) = R\cdot q_1 + L\cdot\dot{q}_1 \quad \Rightarrow \frac{F(t)}{Ap} - \rho g h = R\cdot q_1 + L\cdot\dot{q}_1$$

$$\Rightarrow \dot{x}_1 = \frac{u_1}{ApL} - \frac{\rho g}{L}x_2 - \frac{\rho g}{L}u_2 - \frac{R}{L}x_1$$

h deve essere tutto relativamente allo altro pressione nella formula (rispetto a P_1 è $\rho g(h+d)$, rispetto a P_2 è solo $\rho g h$

$$P_2(t) = +\frac{F_2}{Ap}$$

$$P(t) - P_2(t) = R\cdot q_2 + L\cdot\dot{q}_2$$

$$\rho g h \quad \frac{F_2}{Ap} = R\cdot q_2 + L\cdot\dot{q}_2 \qquad q_2 = Ap\cdot\dot{s}$$

$$\Rightarrow \rho g h - \frac{F_2}{Ap} = R\cdot Ap\cdot\dot{s} + L\cdot Ap\cdot\ddot{s}$$

$$\Rightarrow F_2 = +\rho g h Ap - R\cdot Ap^2\cdot\dot{s} - L\cdot Ap^2\cdot\ddot{s}$$

$$A_s\dot{h} = q_1 - q_2 \Rightarrow \dot{h} = \frac{q_1}{A_s} - \frac{Ap\dot{s}}{A_s}$$

$$m\cdot\ddot{s} = F_2 - K\cdot s - b\cdot\dot{s} \Rightarrow m\ddot{s} = \rho g h Ap - R\cdot Ap^2\dot{s} - LAp^2\ddot{s} - Ks - b\dot{s}$$

$$\Rightarrow (m + LAp^2)\ddot{s} = \rho g h Ap - RAp\dot{s} - Ks - b\dot{s}$$

$$\begin{cases} \dot{x}_1 = \frac{u_1}{ApL} - \frac{\rho g}{L}x_2 - \frac{R}{L}x_1 \\ \dot{x}_2 = \frac{x_1}{A_s} - \frac{Ap}{A_s}x_4 \\ \dot{x}_3 = x_4 \\ \dot{x}_4 = \frac{\rho g Ap}{(m+LAp^2)}x_2 - \frac{RAp}{m+LAp^2}x_4 - \frac{K}{m+LAp^2}x_3 - \frac{b}{m+LAp^2}x_4 \end{cases}$$

$$A = \begin{bmatrix} -\frac{R}{L} & -\frac{\rho g}{L} & 0 & 0 \\ \frac{1}{As} & 0 & 0 & -\frac{Ap}{As} \\ 0 & 0 & 0 & 1 \\ & \frac{\rho g Ap}{(m+LAp^2)} & -\frac{K}{m+LAp^2} & \frac{-RAp-b}{m+LAp^2} \end{bmatrix} \qquad B = \begin{bmatrix} \frac{1}{ApL} & -\frac{\rho g}{L} \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \qquad D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

④ $x(K+1) = Ax(K) + B\cdot u(K)$
$y(K) = C\cdot x(K)$

$$A = \begin{bmatrix} 0.5 & 0.9 \\ 0 & 0.8 \end{bmatrix} \qquad B = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \qquad C = [\,2 \ \ -2\,]$$

- $X(0) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$$\begin{cases} x_1(K+1) = 0.5x_1 + 0.9x_2 + u \\ x_2(K+1) = 0.8x_2 - u \end{cases}$$

$$\begin{cases} x_1(1) = 1 + 0.9 + u \\ x_2(1) = 0.8 - u \end{cases}$$

$$x(K) = A^K\cdot x_0$$

$$\Rightarrow \begin{bmatrix} 0.5 & 0.9 \\ 0 & 0.8 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} & ? & \end{bmatrix}\begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\det(\lambda I - A) = \det\left|\begin{bmatrix} \lambda - 0.5 & 0.9 \\ 0 & \lambda - 0.8 \end{bmatrix}\right| \Rightarrow \lambda^2 - 0.8\lambda - 0.5\lambda + 0.4$$

$$\Rightarrow \lambda^2 - \frac{13}{10}\lambda + \frac{2}{5} \Rightarrow \lambda^2 - \frac{13}{10}\lambda + \frac{2}{5} = 0 \quad \frac{\frac{13}{10} \pm \sqrt{\frac{169}{100} - 4\cdot\frac{2}{5}}}{2}$$

$$= \frac{\frac{13}{10} \pm \sqrt{\frac{9}{100}}}{2} = \frac{\left(\frac{13}{10} \pm \frac{3}{10}\right)}{2} \begin{cases} \frac{1}{2} \\ \frac{4}{5} \end{cases} \qquad \lambda_1 = \frac{1}{2} \quad \lambda_2 = \frac{4}{5}$$

λ_1: 0.5 $\quad \lambda_2$: 0.8

Per λ_1: $(A - 0.5I)V_1 = \bar{0}$

$$\Rightarrow \begin{bmatrix} 0 & 0.9 \\ 0 & 0.3 \end{bmatrix}\begin{Bmatrix} V_{11} \\ V_{12} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \Rightarrow \begin{matrix} 0.9V_{12} = 0 \\ 0.3\cdot V_{12} = 0 \end{matrix} \Rightarrow V_{12} = 0$$

V_{11} è libero ⇒ scegliamo = 1 $\qquad \Rightarrow V_1 = \begin{Bmatrix} 1 \\ 0 \end{Bmatrix}$

per λ_2: $(A - 0.8)V_2 = \bar{0}$

$$\Rightarrow \begin{bmatrix} -0.3 & 0.9 \\ 0 & 0 \end{bmatrix}\bar{V}_2 = \bar{0} \qquad -0.3V_{21} + 0.9V_{22} = 0 \Rightarrow V_{21} = \frac{0.9}{0.3}V_{22}$$

$$\Rightarrow \frac{V_{21}}{V_{22}} = \frac{0.9}{0.3} \qquad \text{scegliamo } V_{21} = 3 \ \ V_{22} = 1$$

$$\Rightarrow V_2 = \begin{Bmatrix} 3 \\ 1 \end{Bmatrix}$$

⇒ matrice di vettori: $\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$ $\qquad \Lambda$ = matrice diagonale di A = $\begin{bmatrix} 0.5 & 0 \\ 0 & 0.8 \end{bmatrix}$

$$P^{-1} = \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix}$$

calcoli: $\frac{\text{adj}(\lambda I - P)}{\det(\lambda I - P)}$

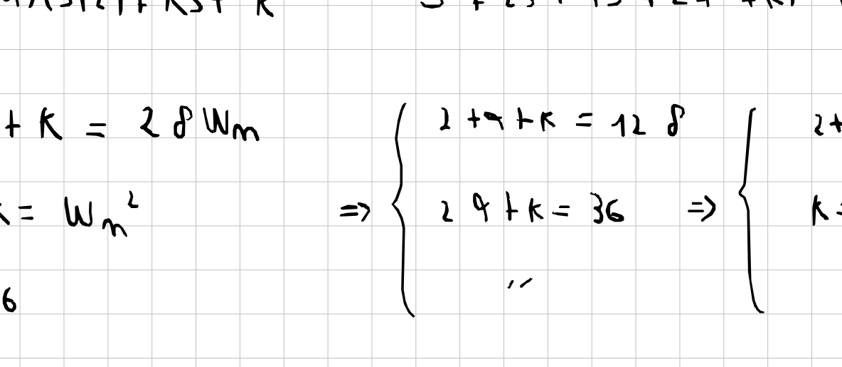
$$A^K = P\Lambda^K P^{-1} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 0.5^K & 0 \\ 0 & 0.8^K \end{bmatrix}\begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.5^K & 3\cdot 0.8^K - 3\cdot 0.5^K \\ 0 & 0.8^K \end{bmatrix}$$

- $x(K) = A^K x(0) \qquad x(2) \equiv K = 2 \quad \Rightarrow x(2) = A^2 x(0)$

$$\Rightarrow x(2) = \begin{bmatrix} 1.67 \\ 0.64 \end{bmatrix} \qquad \begin{bmatrix} 0.5^2 & 3\cdot 0.8^2 - 3\cdot 0.5^2 \\ 0 & 0.8^2 \end{bmatrix}$$

- Altri punti evitati perchè probabilmente non fanno parte del programma

⑤



$$R(s) = \frac{K}{s+a} \qquad G(s) = \frac{s+1}{s+2}$$

- Considerando $d(s) = 0$

$$F(s) = R(s)\cdot G(s) = \frac{K(s+1)}{(s+a)(s+2)}$$

$$\text{Retroazione: } L(s) = \frac{F(s)}{1+F(s)} = \frac{\frac{K(s+1)}{(s+a)(s+2)}}{1 + \frac{K(s+1)}{(s+a)(s+2)}} = \frac{\frac{K(s+1)}{(s+a)(s+2)}}{\frac{(s+a)(s+2) + K(s+1)}{(s+a)(s+2)}}$$

$$= \frac{K(s+1)}{(s+a)(s+2) + Ks + K} = \frac{K(s+1)}{s^2 + 2s + as + 2a + Ks + K} = \frac{K(s+1)}{s^2 + (2+a+K)s + (2a+K)}$$

$$\begin{cases} 2 + a + K = 2\delta\omega_n \\ 2a + K = \omega_n^2 \\ \omega_n = 6 \end{cases} \Rightarrow \begin{cases} 2 + a + K = 12\delta \\ 2a + K = 36 \\ '' \end{cases} \Rightarrow \begin{cases} 2 + a + K = 12\delta \\ K = 36 - 2a \\ '' \end{cases}$$

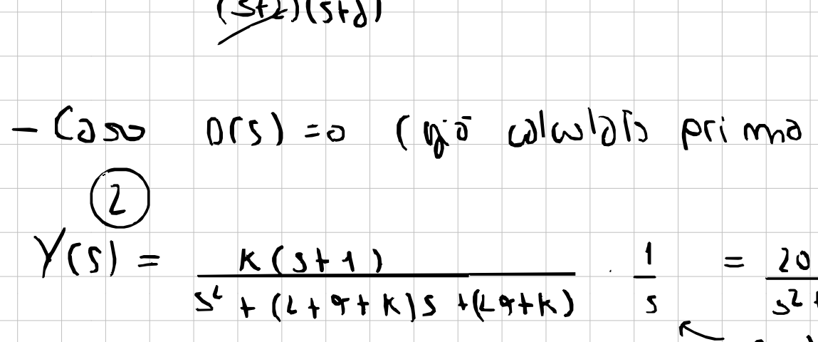
$$T_{as} = \frac{3}{\delta\omega_n} \Rightarrow 0.2 = \frac{\cancel{3}\,1}{\delta\cdot\cancel{6}\,2} \Rightarrow \delta = 2.5$$

$$\Rightarrow \begin{cases} 2 + a + K = \overset{6}{\cancel{12}}\cdot\frac{5}{\cancel{2}} \ (= 2.5) \\ K = 36 - 2a \\ '' \end{cases} \Rightarrow \begin{cases} 2 + a + 36 - 2a = 30 \\ '' \end{cases} \Rightarrow \begin{cases} -a = -8 \\ '' \end{cases} \Rightarrow \begin{cases} a = 8 \\ K = 20 \end{cases}$$

$$\Rightarrow G(s) = \frac{s+1}{s+2} \qquad R(s) = \frac{20}{s+8} \qquad y(\infty) = ?$$

$$y(\infty) = \lim_{t\to\infty} y(t) = \lim_{s\to 0} s\cdot Y(s)$$

- Risposta a r(s) nullo: $E(s) = R(s) - Y(s) = -Y(s)$



$$A'(s) = E(s)\cdot R(s) = R(s)\cdot(-Y(s))$$

$$A(s) = A'(s) + D(s) = R(s)\cdot(-Y(s)) + D(s)$$

$$Y(s) = G(s)\cdot A(s) = G(s)\cdot R(s)(-Y(s)) + G(s)\cdot D(s)$$

$$\Rightarrow Y(s) + G(s)R(s)(Y(s)) = G(s)\cdot D(s) \Rightarrow Y(s)[1 + G(s)R(s)] = G(s)\cdot D(s)$$

$$\Rightarrow Y(s) = \frac{G(s)}{1 + G(s)R(s)}\cdot D(s) \Rightarrow Y(s) = \frac{\frac{s+1}{s+2}}{1 + \frac{20(s+1)}{(s+2)(s+8)}}\cdot\frac{1}{s}$$

D(s) gradino unitario

$$\Rightarrow Y(s) = \frac{\frac{s+1}{\cancel{s+2}}}{\frac{(s+2)(s+8) + 20(s+1)}{\cancel{(s+2)}(s+8)}}\cdot\frac{1}{s} \Rightarrow Y(s) = \frac{(s+1)\cdot(s+8)}{(s+2)(s+8) + 20(s+1)}\cdot\frac{1}{s} \quad ①$$

- Caso $D(s) = 0$ (già calcolato prima)

②
$$Y(s) = \frac{K(s+1)}{s^2 + (2+a+K)s + (2a+K)}\cdot\frac{1}{s} = \frac{20(s+1)}{s^2 + (2+8+20)s + (2\cdot 8+20)}\cdot\frac{1}{s}$$

r(s) gradino unitario

Per principio di sovrapposizione degli effetti $Y(s) = ① + ②$

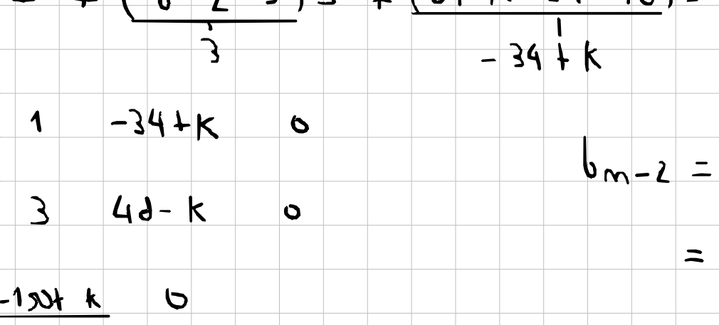
$$\Rightarrow Y(s) = \frac{(s+1)(s+8)}{(s+2)(s+8) + 20(s+1)}\cdot\frac{1}{s} + \frac{20(s+1)}{s^2 + (2+8+20)s + (2\cdot 8+20)}\cdot\frac{1}{s}$$

$$= \frac{1}{s}\cdot\left(\frac{(s+1)(s+8)}{(s+2)(s+8) + 20(s+1)} + \frac{20(s+1)}{s^2 + (2+8+20)s + (2\cdot 8+20)}\right)$$

$$y(\infty) = \lim_{t\to\infty} y(t) = \lim_{s\to 0} sY(s) = \lim_{s\to 0} \cancel{s}\cdot\frac{1}{\cancel{s}}\cdot\left(\frac{(s+1)(s+8)}{(s+2)(s+8) + 20(s+1)} + \frac{20(s+1)}{s^2 + (2+8+20)s + (2\cdot 8+20)}\right)$$

$$= \lim_{s\to 0}\left(\frac{(s+1)(s+8)}{(s+2)(s+8) + 20(s+1)} + \frac{20(s+1)}{s^2 + (2+8+20)s + (2\cdot 8+20)}\right) = \frac{8}{36} + \frac{20}{36} = \frac{\overset{14\,7}{\cancel{28}}}{\cancel{36}_{\,18\,9}} = \boxed{\frac{7}{9}}$$

⑥



$$G(s) = \frac{s-1}{(s+8)(s-2)(s-3)}$$

$$F(s) = G(s)\cdot K = \frac{K(s-1)}{(s+8)(s-2)(s-3)}$$

USA QUESTA RELAZIONE PER LA STABILITÀ

I poli di un sistema in retroazione sono le radici di $1 + G_1G_2 = 0$

$\equiv 1$ in questo caso

$$\Rightarrow 1 + F(s) = 0 \Rightarrow \frac{K(s-1) + (s+8)(s-2)(s-3)}{(s+8)(s-2)(s-3)} = 0$$

$$\Rightarrow K(s-1) + (s+8)(s-2)(s-3) = 0 \Rightarrow Ks - K + (s^2 - 2s + 8s - 16)(s-3)$$

$$\Rightarrow \underline{Ks} - K + s^3 - 2s^2 + 8s^2 \underline{- 16s} - 3s^2 \underline{+ 6s} - \underline{24s} + 48 = 0$$

$$\Rightarrow s^3 + \underbrace{(8-2-3)}_{3}s^2 + \underbrace{(6 + K - 24 - 16)}_{-34+K}s + (48 - K) = 0$$

$$\begin{array}{c|ccc} 3 & 1 & -34+K & 0 \\ 2 & 3 & 48-K & 0 \\ 1 & \frac{-150+4K}{3} & 0 & \\ 0 & 48-K & & \end{array}$$

$$b_{n-2} = \frac{3\cdot(-34+K) - 48 + K}{3} = \frac{-102 + 3K - 48 + K}{3} = \frac{-150 + 4K}{3}$$

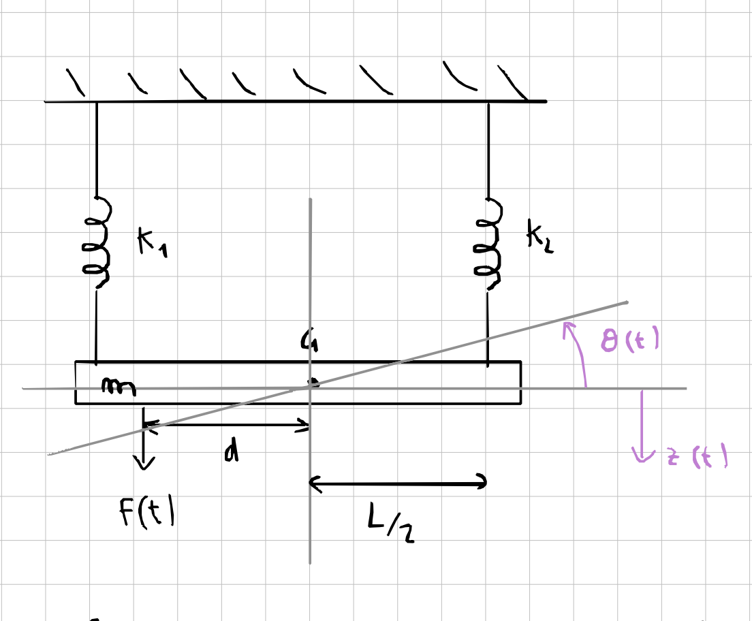
$$c_{n-3} = \frac{\cancel{\left(\frac{-150+4K}{3}\right)}^{1}\cdot(48-K)}{\cancel{\left(\frac{-150+4K}{3}\right)}} = 48 - K$$

$$\Rightarrow \frac{-150+4K}{3} > 0$$

$$48 - 4K > 0$$

$$\Rightarrow \begin{cases} 4K > 150 \\ '' \end{cases} \Rightarrow \begin{cases} K > 37.5 \\ -K > -48 \end{cases} \Rightarrow \boxed{37.5 < K < 48}$$

Prova 2



$$J_G = \frac{ml^2}{3}$$

$$T = \frac{1}{2} m v_G^2 + \frac{1}{2} \cdot \frac{ml^2}{3} \cdot \dot{\theta}^2 = \frac{1}{2} m \left(\dot{z}^2 + \frac{\dot{\theta}^2}{3} l^2 \right)$$

$$U_p = -mgz \qquad U_{E1} = \frac{1}{2} \cdot k_1 \left(z + \frac{l}{2} \cdot \sin\theta \right)^2$$

$$U_{E2} = \frac{1}{2} k_2 \left(z - \frac{l}{2} \sin\theta \right)^2$$

$$U_{TOT} = -mgz + \frac{1}{2} k_1 \left(z + \frac{l}{2} \sin\theta \right)^2 + \frac{1}{2} k_2 \left(z - \frac{l}{2} \sin\theta \right)^2$$

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{\theta}}\right) = \frac{d}{dt}\left(\frac{1}{\cancel{2}} \cdot \frac{ml^2}{3} \cdot \cancel{2}\dot{\theta} \right) = \frac{ml^2}{3} \ddot{\theta} \qquad \frac{\partial T}{\partial \theta} = 0$$

$$\frac{\partial U}{\partial \theta} = \frac{1}{2} k_1 \left[2 \left(z + \frac{l}{2} \sin\theta \right) \cdot \frac{l}{\cancel{2}} \cos\theta \right] + \frac{1}{2} k_2 \left[\cancel{2} \left(z - \frac{l}{2} \sin\theta \right) \cdot \left(-\frac{l}{2} \cos\theta \right) \right]$$

$$= \frac{1}{2} k_1 l z \cos\theta + \frac{l^2}{4} k_1 \sin\theta \cos\theta - \frac{1}{2} k_2 l z \cos\theta + \frac{l^2}{4} k_2 \sin\theta \cos\theta$$

$$= \frac{l}{2} z \cos\theta \, (k_1 - k_2) + \frac{l^2}{4} \sin\theta \cos\theta \, (k_1 + k_2)$$

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{z}}\right) = \frac{d}{dt}\left(\frac{1}{\cancel{2}} \cdot \cancel{2} m \dot{z} \right) = m \ddot{z} \qquad \frac{\partial T}{\partial z} = 0$$

$$\frac{\partial U}{\partial z} = -mg + \frac{1}{\cancel{2}} k_1 \, \cancel{2} \left(z + \frac{l}{2} \sin\theta \right) + \frac{1}{\cancel{2}} k_2 \, \cancel{2} \left(z - \frac{l}{2} \sin\theta \right)$$

$$= -mg + k_1 z + \frac{k_1 l}{2} \sin\theta + k_2 z - \frac{l}{2} k_2 \sin\theta$$

$$= -mg + z(k_1 + k_2) + \frac{l}{2} \sin\theta \, (k_1 - k_2)$$

spostamento causato da F (d è la distanza da G)

$$\delta L = F \cdot \delta x = F \cdot \delta(z + d \sin\theta)$$

$$= F \cdot \frac{\partial(z + d \sin\theta)}{\partial z} \delta z + F \cdot \frac{\partial(z + d \cdot \sin\theta)}{\partial \theta} \cdot \delta\theta = F \cdot \delta z + F \cdot d \cos\theta \, \delta\theta$$

$$= F \cdot \delta z + F \cdot d \cdot \cos\theta \, \delta\theta \qquad F \cdot \delta z + F \cdot d \cdot \cos\theta \, \delta\theta = Q_1 \cdot \delta z + Q_2 \cdot \delta\theta \Rightarrow \begin{matrix} Q_1 = F \\ Q_2 = F \cdot d \cdot \cos\theta \end{matrix}$$

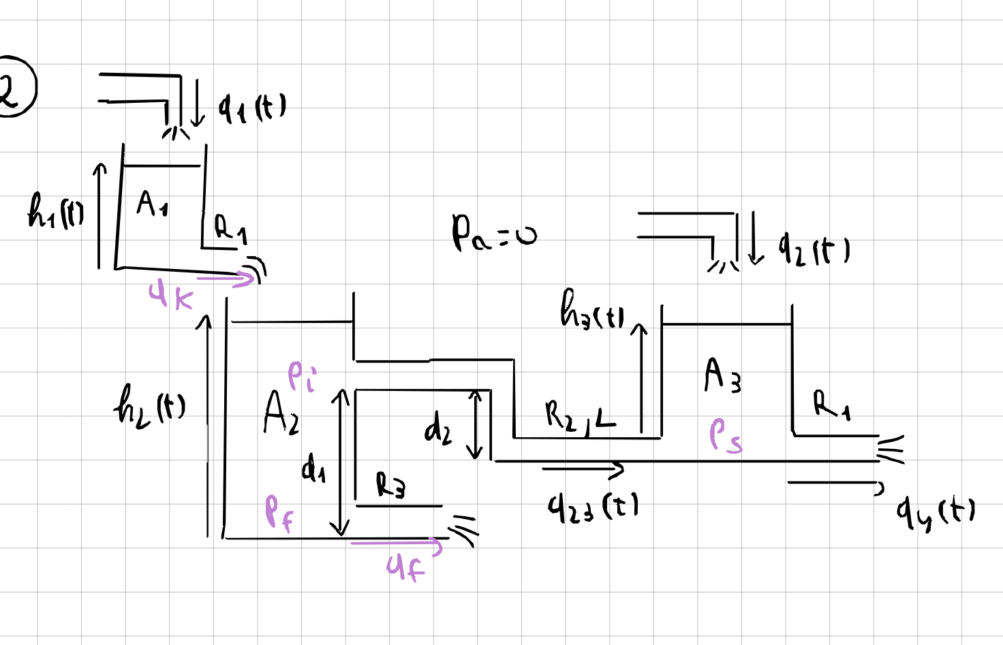
$$\Rightarrow \begin{cases} \dfrac{ml^2}{3} \ddot{\theta} + \dfrac{l}{2} z \cos\theta (k_1 - k_2) + \dfrac{l^2}{4} \sin\theta \cos\theta (k_1 + k_2) = F \cdot d \cdot \cos\theta \\ m\ddot{z} - mg + z(k_1 + k_2) + \dfrac{l}{2} \sin\theta (k_1 - k_2) = F \end{cases}$$

Scegliendo $x_1 = z \quad x_2 = \dot{z} \quad x_3 = \theta \quad x_4 = \dot{\theta} \quad u = F$

$$\begin{cases} \dfrac{ml^2}{3} \dot{x}_4 + \dfrac{l}{2} x_1 \cos x_3 (k_1 - k_2) + \dfrac{l^2}{4} \sin x_3 \cos x_3 (k_1 + k_2) = u \cdot d \cdot \cos x_3 \\ m\ddot{z} - mg + x_1(k_1 + k_2) + \dfrac{l}{2} \sin x_3 (k_1 - k_2) = u \end{cases}$$

$$\Rightarrow \begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = \dfrac{u}{m} + g - \dfrac{x_1}{m}(k_1 + k_2) - \dfrac{l}{2m} \sin x_3 (k_1 - k_2) \\ \dot{x}_3 = x_4 \\ \dot{x}_4 = \dfrac{3ud}{ml^2} \cos x_3 - \dfrac{3}{2ml} x_1 \cos x_3 (k_1 - k_2) - \dfrac{3}{4m} \sin x_3 \cos x_3 (k_1 + k_2) \end{cases}$$

②



$u_1 = q_1 \quad u_2 = q_2 \quad u_3 = d_1 \quad u_4 = d_2 \qquad y_1 = h_1 \quad y_2 = h_3 \quad y_3 = q_y(t)$

$$A_1 \cdot \dot{h}_1 = q_1 - q_k$$

$$\rho g h - P_a = R_1 \cdot q_k \Rightarrow q_k = \frac{\rho g h}{R_1}$$

$$\Rightarrow A_1 \cdot \dot{h}_1 = q_1 - \frac{\rho g h}{R_1}$$

$$P_i = \rho \cdot g \cdot (h_2 - d_1)$$

$$A_2 \cdot \dot{h}_2 = \frac{\rho g h_1}{R_1} - q_{23} - q_f \qquad \rho g h - 0 = R_3 q_f \Rightarrow q_f = \frac{\rho g h_2}{R_3}$$

$$P_i - P_s = R_2 \cdot q_{23} + L \cdot \dot{q}_{23}$$

$$\rho g (h_2 - d_1) - \rho g (h_3 - d_2) = R_2 \cdot q_{23} + L \cdot \dot{q}_{23}$$

$$A_3 \cdot \dot{h}_3 = q_{23} + q_2 - q_y \qquad P_s - P_A = R_1 \cdot q_y \Rightarrow q_y = \frac{\rho \cdot g \cdot h_3}{R_1}$$

$x_1 = h_1, \; x_2 = h_2, \; x_3 = q_{23}, \; x_4 = h_3$

$$\dot{h}_1 = \frac{q_1}{A_1} - \frac{\rho g h_1}{A_1 R_1}$$

$$\dot{h}_2 = \frac{\rho g h_1}{A_2 R_1} - \frac{q_{23}}{A_2} - \frac{\rho g h_2}{A_2 R_3}$$

$$\dot{q}_{23} = \frac{\rho g (h_2 - d_1)}{L} - \frac{\rho g (h_3 - d_2)}{L} - \frac{R_2 q_{23}}{L}$$

$$\dot{h}_3 = \frac{q_{23}}{A_3} + \frac{q_2}{A_3} - \frac{\rho g h_3}{A_3 R_1}$$

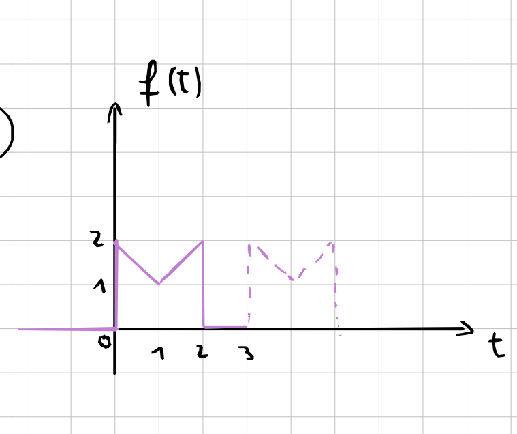
$$\begin{cases} \dot{x}_1 = \dfrac{u_1}{A_1} - \dfrac{\rho g}{A_1 R_1} \cdot x_1 \\ \dot{x}_2 = \dfrac{\rho g}{A_2 R_1} \cdot x_1 - \dfrac{x_3}{A_2} - \dfrac{\rho g}{A_2 R_3} \cdot x_2 \\ \dot{x}_3 = \dfrac{\rho g}{L} x_2 - \dfrac{\rho g}{L} u_3 - \dfrac{\rho g}{L} x_4 + \dfrac{\rho g}{L} u_4 - \dfrac{R_2}{L} x_3 \\ \dot{x}_4 = \dfrac{x_3}{A_3} + \dfrac{u_2}{A_3} - \dfrac{\rho g}{R_4 A_3} \cdot x_4 \end{cases}$$

$$A = \begin{bmatrix} -\frac{\rho g}{A_1 R_1} & 0 & 0 & 0 \\ \frac{\rho g}{A_2 R_1} & -\frac{\rho g}{A_2 R_3} & -\frac{1}{A_2} & 0 \\ 0 & \frac{\rho g}{L} & -\frac{R_2}{L} & -\frac{\rho g}{L} \\ 0 & 0 & \frac{1}{A_3} & -\frac{\rho g}{R_4 A_3} \end{bmatrix} \qquad B = \begin{bmatrix} \frac{1}{A_1} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\rho g}{L} & \frac{\rho g}{L} \\ 0 & \frac{1}{A_3} & 0 & 0 \end{bmatrix}$$

$$y_1 = h_1 \qquad y_2 = h_3 \qquad y_3 = \frac{\rho g h_3}{R_1}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & \frac{\rho g}{R_4} \end{bmatrix} \qquad D = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

③



T = 3s

Non scrivere come li fai i passaggi

$t=0 \Rightarrow$ salto ampiezza 2 $\Rightarrow 2(u(t))$ + rampa $-r(t)$

$\Rightarrow 2(u(t)) - r(t)$

$t=1 \Rightarrow$ precedente: rampa con $m=-1$ $\Rightarrow$ successivo: rampa con $m=1$

$\Rightarrow m = -1 - (1) = -2 \Rightarrow -2(t-1)r(t-1)$

$t=2 \Rightarrow$ c'è un balzo negativo di ampiezza 2 $\Rightarrow -2u(t-2)$

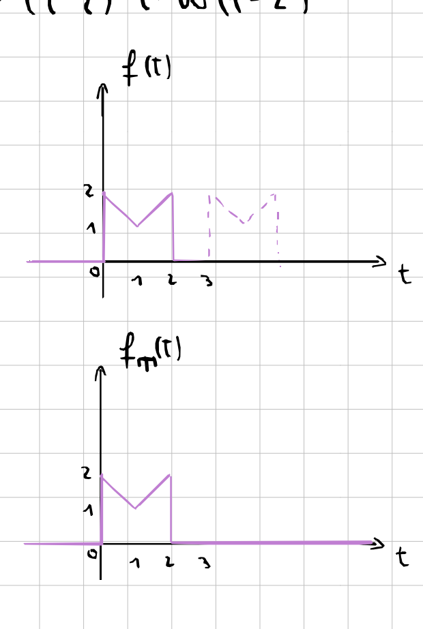
m prec. = 1 m succ = 0 $\Rightarrow m = 0 - 1 = -1 \Rightarrow$ + rampa $\underset{\uparrow -1}{-}r(t-2)$

$\Rightarrow -2u(t-2) - r(t-2)$

$$\Rightarrow f(t) = 2u(t) - r(t) - 2(t-1)r(t-1) - 2u(t-2) - r(t-2)$$

$$r(t) = t\,u(t) \Rightarrow 2u(t) - t\,u(t) - 2(t-1)\,t\,u(t) - 2u(t-2) - (t-2) \cdot t \cdot u(t-2)$$

$$F(s) = \frac{F_T(s)}{1 - e^{-Ts}} \qquad f_T(t) = \begin{cases} f(t) & 0 \le t \le T \\ 0 & t > T \end{cases}$$



$$F_T(s) = \frac{2}{s} - \frac{1}{s^2} - \frac{2}{s^2} e^{-s} - \frac{2}{s} \cdot e^{-2s} - \frac{1}{s^2} \cdot e^{-2s}$$

$$\Rightarrow F(s) = \frac{\frac{2}{s} - \frac{1}{s^2} - \frac{2}{s^2} e^{-s} - \frac{2}{s} \cdot e^{-2s} - \frac{1}{s^2} \cdot e^{-2s}}{1 - e^{-3s}}$$

④

$$\begin{cases} \dot{x}_1(t) = x_2(t) - x_1(t) \\ \dot{x}_2(t) = -x_1(t) + 9x_2(t) - 2x_2^3(t) \end{cases}$$

- stat. di equilibrio

$$\begin{cases} \dot{x}_1 = 0 \\ \dot{x}_2 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = x_2 \\ -x_2 + 9x_2 - 2x_2^3 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = x_2 \\ 2x_2(4x_2 - x_2^2) = 0 \end{cases}$$

$$x_e' = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \qquad x_e'' = \begin{Bmatrix} 2 \\ 2 \end{Bmatrix} \qquad x_e''' = \begin{Bmatrix} -2 \\ -2 \end{Bmatrix}$$

- stabilità

$$f_1 = x_2 - x_1 \qquad f_2 = 8x_2 - 2x_2^3$$

$$\frac{\partial f_1}{\partial x_1} = -1 \qquad \frac{\partial f_1}{\partial x_2} = 1 \qquad \frac{\partial f_2}{\partial x_1} = 0 \qquad \frac{\partial f_2}{\partial x_2} = 8 - 6x_2^2$$

$$A = \begin{bmatrix} -1 & 1 \\ 0 & 8 - 6x_2^2 \end{bmatrix} \qquad \lambda I - A = \begin{bmatrix} \lambda + 1 & 1 \\ 0 & \lambda - 8 + 6x_2^2 \end{bmatrix}$$

$$\det(sI - A) = \lambda^2 - 8\lambda + 6x_2^2 \lambda + \lambda - 8 + 6x_2^2 = \lambda^2 + (6x_2^2 - 8 + 1)\lambda - 8 + 6x_2^2$$

$$= \lambda^2 + (6x_2^2 - 7)\lambda - 8 + 6x_2^2$$

$$\text{in } \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \Rightarrow \lambda^2 - 7\lambda - 8 = 0 \qquad \frac{7 \pm \sqrt{49 - 4(-8)}}{2} = \frac{7 \pm 9}{2} \begin{matrix} \nearrow -1 \\ \searrow 8 \end{matrix}$$

$\Rightarrow 0,0$ è instabile

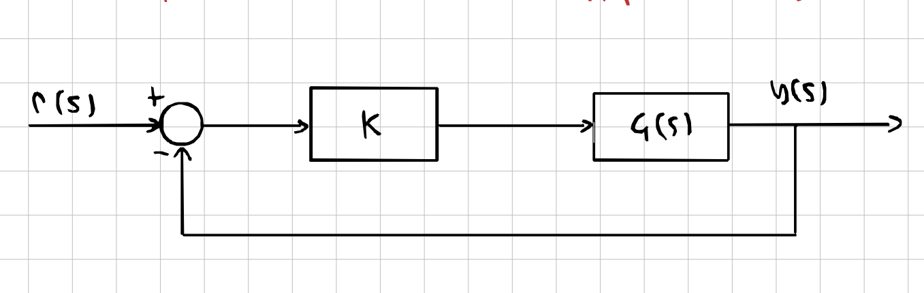
$$\text{in } \begin{Bmatrix} 2 \\ 2 \end{Bmatrix} \Rightarrow \lambda^2 + (6 \cdot 4 - 7)\lambda - 8 + 6 \cdot 4 = \lambda^2 + 17\lambda + 16 = 0$$

$n = 2$, tutti i coefficienti hanno lo stesso segno $\Rightarrow$ tutti i λ hanno parte reale negativa $\Rightarrow \begin{Bmatrix} 2 \\ 2 \end{Bmatrix} \Rightarrow$ asintoticamente stabile

in $\begin{Bmatrix} -2 \\ -2 \end{Bmatrix}$ è analogo a $\begin{Bmatrix} 2 \\ 2 \end{Bmatrix} \Rightarrow$ asintoticamente stabile

⑤ $\Rightarrow$ da capire se i sistemi a tempo discreto sono oggetto d'esame

⑥



$$G(s) = \frac{s+1}{s^2(s^2 + 6s + 25)} \qquad L(s) = \frac{K \cdot (s+1)}{s^2(s^2 + 6s + 25)}$$

Per un sistema in retroazione, $1 + L(s) = 0 \Rightarrow 1 + \dfrac{K(s+1)}{s^2(s^2+6s+25)} = 0$

$$\Rightarrow \frac{K(s+1) + s^2(s^2 + 6s + 25)}{s^2(s^2 + 6s + 25)} = 0 \Rightarrow K(s+1) + s^2(s^2 + 6s + 25) = 0$$

$$s^4 + 6s^3 + 25s^2 + Ks + K = 0$$

$$\underset{n^\circ \nearrow \; \nwarrow \text{colonna}}{X_{i,h}} = -\frac{1}{X_{i+1,1}} \cdot \det \left| \begin{bmatrix} X_{i+2,1} & X_{i+2,h+1} \\ X_{i+1,1} & X_{i+1,h+1} \end{bmatrix} \right|$$

$$\begin{array}{c|ccc} 4 & 1 & 25 & K \\ 3 & 6 & K & 0 \\ 2 & \frac{150-K}{6} & K & \\ 1 & \frac{+114K - K^2}{150-K} & & \\ 0 & K & & \end{array}$$

$$X_{2,1} = -\frac{1}{6} \cdot \det \left[\begin{pmatrix} 1 & 25 \\ 6 & K \end{pmatrix} \right] = -\frac{K - 150}{6} = \frac{150 - K}{6}$$

$$X_{2,2} = -\frac{1}{6} \cdot \det \left[\begin{pmatrix} 1 & K \\ 6 & 0 \end{pmatrix} \right] = -\frac{-6K}{6} = K$$

$$X_{1,1} = -\frac{6}{150 - K} \cdot \det \left[\begin{pmatrix} 6 & K \\ \frac{150-K}{6} & K \end{pmatrix} \right] = \frac{-6 \cdot \left(6K - \left(\frac{150-K}{6}\right) K\right)}{150 - K}$$

$$= \frac{-6 \cdot K \left(6 - \frac{150 - K}{6}\right)}{150 - K} = \frac{-\cancel{6}K \left(\frac{36 - 150 + K}{\cancel{6}}\right)}{150 - K} = \frac{+114K - K^2}{150 - K}$$

$$X_{0,1} = -\frac{150 - K}{114K - K^2} \cdot \det \left| \begin{bmatrix} \frac{150-K}{6} & K \\ \frac{+114K - K^2}{150-K} & 0 \end{bmatrix} \right| = +K$$

$$\Rightarrow \begin{cases} \dfrac{150 - K}{6} > 0 \\ \dfrac{114K - K^2}{150 - K} > 0 \\ K > 0 \end{cases} \Rightarrow \begin{cases} K < 150 \\ K(114 - K) > 0 \; \begin{cases} K > 0 \\ K < 114 \end{cases} \\ K > 0 \end{cases}$$

$$\Rightarrow 0 < K < 114$$

- $K(114 - K) = 0 \Rightarrow K = 114$

$\rightarrow$ riga sopra $\Rightarrow \left(\dfrac{150 - K}{6}\right)s^2 + K = 0 \Rightarrow (150 - K)s^2 + 6K = 0$

$$\Rightarrow s^2 = \frac{-6K}{150 - K} \Rightarrow K = 114 \Rightarrow \frac{-6 \cdot 114}{150 - 114} = \frac{-\cancel{6} \cdot \overset{57}{\cancel{114}}}{\cancel{36}_{6}} = -\frac{57}{3}$$

poli: $p_{1,2} = \pm j\sqrt{\dfrac{57}{3}} = \pm j\sqrt{19}$

⑦

$$\begin{cases} \dot{x}(t) = A\,x(t) \\ y(t) = B\,u(t) \end{cases} \qquad A = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} \qquad C = [1 \quad 0]$$

andrebbe s al posto di λ

$$y_l(t) = C \cdot e^{At} x_0$$

$$e^{At} = \mathcal{L}^{-1}[(sI - A)^{-1}] \qquad sI - A = \begin{bmatrix} \lambda - 2 & +1 \\ -3 & \lambda + 2 \end{bmatrix}$$

$$\det(sI - A) = (\lambda - 2)(\lambda + 2) + 3 \Rightarrow \lambda^2 + 2\lambda - 2\lambda - 4 + 3 = \lambda^2 - 1 = (\lambda - 1)(\lambda + 1)$$

adj $(sI - A)$

$$C_{11} = 1 \cdot (\lambda + 2) = \lambda + 2$$

$$C_{12} = -1 \cdot (-3) = 3 \qquad \Rightarrow \begin{bmatrix} \lambda - 2 & 3 \\ -1 & \lambda + 2 \end{bmatrix}^T = \begin{bmatrix} \lambda + 2 & -1 \\ 3 & \lambda - 2 \end{bmatrix}$$

$$C_{21} = -1 \cdot (1) = -1$$

$$C_{22} = 1 \cdot (\lambda - 2) = \lambda - 2$$

$$\Rightarrow e^{At} = \mathcal{L}^{-1} \begin{bmatrix} \frac{\lambda + 2}{(\lambda - 1)(\lambda + 1)} & \frac{-1}{(\lambda - 1)(\lambda + 1)} \\ \frac{3}{(\lambda - 1)(\lambda + 1)} & \frac{\lambda - 2}{(\lambda - 1)(\lambda + 1)} \end{bmatrix}$$

$$- \; \frac{\lambda - 2}{(\lambda - 1)(\lambda + 1)} \qquad \frac{A}{\lambda - 1} + \frac{B}{\lambda + 1} = \lambda - 2 \Rightarrow A\lambda + A + B\lambda - B = \lambda - 2$$

$$\begin{cases} A + B = 1 \\ A - B = -2 \end{cases} \Rightarrow \begin{cases} A = 1 - B \\ 1 - B - B = -2 \end{cases} \Rightarrow \begin{cases} \text{"} \\ -2B = -3 \end{cases} \Rightarrow \begin{cases} A = \frac{5}{2} \\ B = \frac{3}{2} \end{cases}$$

$$\Rightarrow \frac{\frac{5}{2}}{\lambda - 1} + \frac{\frac{3}{2}}{\lambda + 1} \Rightarrow \frac{5}{2(\lambda - 1)} + \frac{3}{2(\lambda + 1)} \qquad \mathcal{L}^{-1} \Rightarrow \frac{5}{2} e^t + \frac{3}{2} e^{-t}$$

$$\rightarrow \frac{3}{(\lambda - 1)(\lambda + 1)} = A\lambda + A + B\lambda - B = 3 \qquad \begin{cases} A - B = 3 \\ A + B = 0 \end{cases} \Rightarrow \begin{cases} B = -\frac{3}{2} \\ A = -B \end{cases}$$

$$\frac{\frac{3}{2}}{\lambda - 1} - \frac{\frac{3}{2}}{\lambda + 1} \Rightarrow \mathcal{L}^{-1} = \frac{3}{2} e^t - \frac{3}{2} e^{-t}$$

$$\rightarrow \frac{-1}{(\lambda - 1)(\lambda + 1)} = A\lambda + A + B\lambda - B = -1 \Rightarrow \begin{cases} A + B = 0 \\ A - B = -1 \end{cases} \Rightarrow \begin{cases} A = -B \\ B = \frac{1}{2} \end{cases}$$

$$\Rightarrow \mathcal{L}^{-1} \Rightarrow -\frac{1}{2} e^t + \frac{1}{2} e^{-t}$$

$$\rightarrow \frac{\lambda + 2}{(\lambda - 1)(\lambda + 1)} \Rightarrow A\lambda + A + B\lambda - B = \lambda + 2 \qquad \begin{cases} A - B = 2 \\ A + B = 1 \end{cases} \Rightarrow \begin{cases} 1 - B - B = 2 \\ A = 1 - B \end{cases}$$

$$\Rightarrow \begin{cases} B = -\frac{1}{2} \\ A = \frac{3}{2} \end{cases} \Rightarrow \frac{\frac{3}{2}}{\lambda - 1} - \frac{\frac{1}{2}}{\lambda + 1} \Rightarrow \mathcal{L}^{-1} \Rightarrow \frac{3}{2} e^t - \frac{1}{2} e^{-t}$$

$$e^{At} = \begin{bmatrix} \frac{3}{2} e^t - \frac{1}{2} e^{-t} & -\frac{1}{2} e^t + \frac{1}{2} e^{-t} \\ \frac{3}{2} e^t - \frac{3}{2} e^{-t} & \frac{5}{2} e^t + \frac{3}{2} e^{-t} \end{bmatrix}$$

$$\Rightarrow y_l(t) = [1 \quad 0] \begin{bmatrix} \frac{3}{2} e^t - \frac{1}{2} e^{-t} & -\frac{1}{2} e^t + \frac{1}{2} e^{-t} \\ \frac{3}{2} e^t - \frac{3}{2} e^{-t} & \frac{5}{2} e^t + \frac{3}{2} e^{-t} \end{bmatrix} \cdot \begin{Bmatrix} x_0 \\ x_1 \end{Bmatrix}$$

$$= [1 \quad 0] \begin{bmatrix} \left(\frac{3}{2} e^t - \frac{1}{2} e^{-t}\right) x_0 & \left(-\frac{1}{2} e^t + \frac{1}{2} e^{-t}\right) x_1 \\ \left(\frac{3}{2} e^t - \frac{3}{2} e^{-t}\right) x_0 & \left(\frac{5}{2} e^t + \frac{3}{2} e^{-t}\right) x_1 \end{bmatrix}$$

$$y_l(t) = \left(\frac{3}{2} e^t - \frac{1}{2} e^{-t}\right) x_0 + \left(-\frac{1}{2} e^t + \frac{1}{2} e^{-t}\right) x_1$$

$$= \frac{3}{2} e^t x_0 - \frac{1}{2} e^{-t} x_0 - \frac{1}{2} e^t x_1 + \frac{1}{2} e^{-t} x_1 = \frac{1}{2} e^{-t} (x_1 - x_0) + \frac{1}{2} e^t (3x_0 - x_1)$$

Affinché $y_l(t)$ sia asintoticamente stabile, $y(\infty) = 0$

$$\Rightarrow \underbrace{\frac{1}{2} e^{-t} (x_1 - x_0)}_{\text{questo converge sempre}} + \underbrace{\frac{1}{2} e^t (3x_0 - x_1)}_{\text{termine cattivo (va a infinito)}}$$

↳ per far sì che y_l converga, questo deve essere 0 (deve annullarsi)

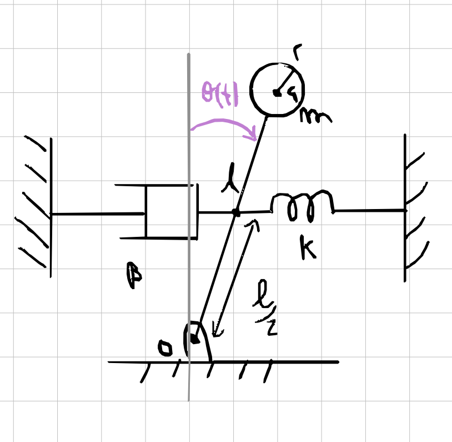
$$\Rightarrow 3x_0 = x_1$$

$$\Downarrow$$

$$x_0 = \begin{Bmatrix} \alpha \\ 3\alpha \end{Bmatrix} \quad \left(\text{o anche } x_0 = \begin{Bmatrix} \alpha/3 \\ \alpha \end{Bmatrix} \right)$$

con $\alpha \in \mathbb{R}$

Prova 5



$$J_G = \frac{2}{5} m r^2$$

$$T = \frac{1}{2} m v^2 + \frac{1}{2} J_G \dot{\theta}^2$$

$$OG = < (l+r)\sin\theta , (l+r)\cos\theta >$$

errore qui, $\frac{d}{dt}\left((l+r)\cos\theta\right)$ $= -(l+r)\dot{\theta}\sin\theta$

$$\Rightarrow v = < (l+r)\cos\theta , -(l+r)\sin\theta >$$

$$|v|^2 = \left(\sqrt{(l+r)^2\cos^2\theta + (l+r)^2\sin^2\theta}\right)^2 = (l+r)^2$$

$$\Rightarrow T = \frac{m(l+r)^2}{2} + \frac{\dot{\theta}^2 m r^2}{5}$$

$$U = U_P + U_E$$

$$U_P = -mg(l+r)\cos\theta \qquad U_E = \frac{1}{2}\cdot K \cdot \left(-\frac{l}{2}\sin\theta\right)^2$$

$$\Rightarrow U = -mg(l+r)\cos\theta - \frac{Kl^2}{8}\sin^2\theta$$

$$R = \frac{1}{2}\beta\left(\frac{d}{dt}\left(\frac{l}{2}\sin\theta\right)\right)^2 = \frac{1}{2}\beta\left(\frac{l}{2}\cdot\cos\theta\cdot\dot{\theta}\right)^2 = \frac{1}{2}\beta\cdot\frac{l^2}{4}\cos^2\theta\,\dot{\theta}^2$$

$$= \frac{\beta l^2}{8}\cos^2\theta\,\dot{\theta}^2$$

$$\frac{d}{dt}\left(\frac{\partial T}{\partial\dot{\theta}}\right) = \frac{2}{5}\ddot{\theta} m r^2 \qquad \frac{\partial T}{\partial\theta} = 0 \qquad \frac{\partial U}{\partial\theta} = mg(l+r)\sin\theta - \frac{Kl^2}{\cancel{8}4}\cdot 2\sin\theta\cdot\cos\theta$$

$$\frac{\partial R}{\partial\dot{\theta}} = \frac{\beta l^2}{4}\cos^2\theta\,\dot{\theta}$$

no ingressi $\Rightarrow Q_1 = 0$ (no forze esterne da considerare)

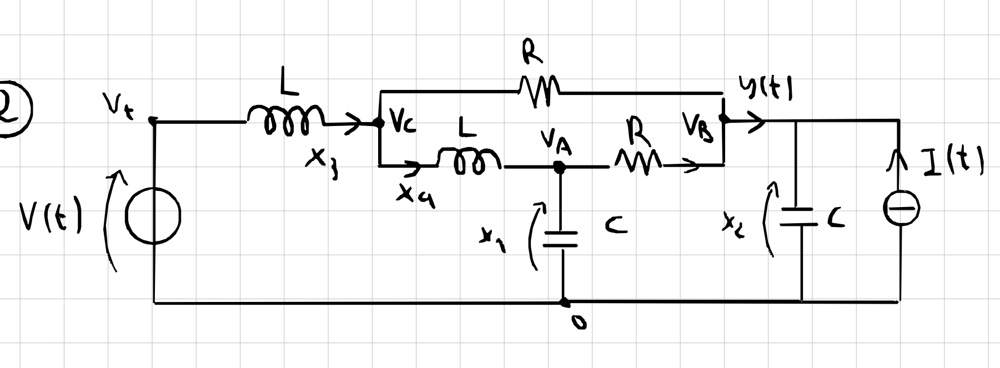
$$\frac{2}{5}\ddot{\theta} m r^2 + mg(l+r)\sin\theta - \frac{Kl^2}{4}\sin\theta\cos\theta + \frac{\beta l^2}{4}\cos^2\theta\,\dot{\theta} = 0$$

$$\ddot{\theta} = -\frac{\cancel{m}g(l+r)\sin\theta}{2\cancel{m}r^2}\cdot 5 + \frac{Kl^2\sin\theta\cos\theta\cdot 5}{8mr^2} - \frac{\beta l^2\cdot 5}{8mr^2}\cos^2\theta\,\dot{\theta}$$

Scegliendo $x_1 = \theta \quad x_2 = \dot{\theta}$

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = 5\cdot\frac{g(l+r)}{2r^2}\sin x_1 + \frac{5}{8}\frac{Kl^2}{mr^2}\sin x_1\cos x_1 - \frac{5}{8}\frac{\beta l^2}{mr^2}\cos^2 x_1\cdot x_2 \end{cases}$$

②



SBAGLIATO

ingressi: $I(t), V(t)$ uscita $y(t)$

$$\text{LKT}_1: \; V(t) - L\dot{x}_3 - L\dot{x}_4 - x_1 = 0$$

$$\left.\begin{array}{l} \text{LKT}_3: \; x_1 - R\cdot(i_1) - x_2 = 0 \\ \text{LKC}_3: \; x_4 - C\dot{x}_1 - i_1 = 0 \end{array}\right\} \Rightarrow x_1 - R(x_4 - C\dot{x}_1) - x_2 = 0$$

$$\text{LKC}_4: \; -C\dot{x}_2 + I(t) - (x_4 - C\dot{x}_1) - \frac{(L\dot{x}_4 + (x_2 - x_1))}{R} = 0$$

$$x_3 - x_4 - \frac{(L\dot{x}_4 + (x_2 - x_1))}{R} = 0$$

$$-\frac{L\dot{x}_4}{R} - \frac{x_2}{R} + \frac{x_1}{R} + x_3 - x_4 = 0 \qquad \dot{x}_4 = -\frac{x_2}{L} + \frac{x_1}{L} + Rx_3 - Rx_4$$

$$V(t) - L\dot{x}_3 - \cancel{L}\left(-\frac{x_2}{\cancel{L}} + \frac{x_1}{\cancel{L}} + Lx_3 - LRx_4\right) = 0 \qquad \Rightarrow \dot{x}_3 = -Rx_3 + Rx_4 + \frac{V(t)}{L}$$

$$-C\dot{x}_2 + I(t) - x_4 + C\cdot\dot{x}_1 - \frac{\left(-x_2 + x_1 + \frac{R}{L}x_3 - \frac{R}{L}x_4\right)}{R} = 0$$

$$\Rightarrow -C\dot{x}_2 + C\dot{x}_1 + I(t) - x_4 - \frac{(-x_2L + x_1L + Rx_3 - Rx_4)}{L\cdot R} = 0$$

$$\Rightarrow C\dot{x}_2 = C\dot{x}_1 + I(t) - x_4 + \frac{x_2}{R} - \frac{x_1}{R} - \frac{x_3}{L} - \frac{x_4}{L}$$

$$x_1 - Rx_4 + CR\dot{x}_1 - x_2 = 0 \quad \Rightarrow \dot{x}_1 = \frac{x_2}{RC} + \frac{x_4}{C} - \frac{x_1}{RC}$$

$$\Rightarrow C\dot{x}_2 = \frac{x_2}{R} + \cancel{x_4} - \frac{x_1}{RC} - \cancel{x_4} + \frac{x_2}{R} - \frac{x_1}{R} - \frac{x_3}{L} - \frac{x_4}{L}$$

$$\Rightarrow \dot{x}_2 = -\frac{x_1}{R} - \frac{x_1}{RC} + \frac{2x_2}{RC} - \frac{x_3}{L} - \frac{x_4}{L}$$

$$\Rightarrow \begin{cases} \dot{x}_1 = -\frac{x_1}{RC} + \frac{x_2}{RC} + \frac{x_4}{C} \\ \dot{x}_2 = -\frac{x_1}{R}\left(1 - \frac{1}{C}\right) + \frac{2}{RC}x_2 - \frac{x_3}{L} - \frac{x_4}{L} \\ \dot{x}_3 = -Rx_3 + Rx_4 + \frac{V(t)}{L} \\ \dot{x}_4 = \frac{x_1}{L} - \frac{x_2}{L} + Rx_3 - Rx_4 \end{cases}$$

$$y = -(x_4 - C\dot{x}_1) - \frac{(L\dot{x}_4 + (x_2 - x_1))}{R}$$

$$= -x_4 - \left(\cancel{C}\left(-\frac{x_1}{R\cancel{C}} + \frac{x_2}{R\cancel{C}} + \frac{x_4}{\cancel{C}}\right)\right) - \frac{L}{R}\cdot\left(\frac{x_1}{L} - \frac{x_2}{L} + Rx_3 - Rx_4\right) - \frac{x_2}{R} + \frac{x_1}{R}$$

$$= -x_4 + x_1 - x_2 - x_4 - \cancel{\frac{x_1}{R}} + \cancel{\frac{x_2}{R}} - Lx_3 + Lx_4 - \cancel{\frac{x_2}{R}} + \cancel{\frac{x_1}{R}}$$

$$= x_1 - x_2 - Lx_3 + x_4(L-2)$$

③

$$\begin{cases} \dot{x}_1(t) = -3x_1(t) + 2x_2^2(t) + u(t) \\ \dot{x}_2(t) = x_1(t) + 2x_2(t) \end{cases}$$

$$\begin{cases} \dot{x}_1 = 0 \\ \dot{x}_2 = 0 \end{cases} \Rightarrow \begin{cases} -3x_1(t) + 2x_2^2(t) + u(t) = 0 \\ x_1(t) + 2x_2(t) = 0 \end{cases} \Rightarrow \text{quando } u = 0$$

$$\Rightarrow \begin{cases} -3x_1 + 2x_2^2 = \\ x_1 + 2x_2 = 0 \end{cases} \Rightarrow \begin{cases} 6x_2 + 2x_2^2 = 0 \\ x_1 = -2x_2 \end{cases} \Rightarrow \begin{cases} 2x_2(3 + x_2) = 0 \\ x_1 = -2x_2 \end{cases}$$

$$\bar{x}' = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad \bar{x}'' = \begin{bmatrix} 6 \\ -3 \end{bmatrix} \leftarrow$$ falli come matrici

$$f_1 = -3x_1 + 2x_2^2 \qquad \frac{\partial f_1}{\partial x_1} = -3 \qquad \frac{\partial f_1}{\partial x_2} = 4x_2$$

$$f_2 = x_1 + 2x_2 \qquad \frac{\partial f_2}{\partial x_1} = 1 \qquad \frac{\partial f_2}{\partial x_2} = 2$$

$$\lambda I - A = \begin{bmatrix} \lambda+3 & -4x_2 \\ -1 & \lambda-2 \end{bmatrix} \qquad \det(sI - A) = (\lambda+3)(\lambda-2) - (-4x_2\cdot(-1))$$

$$= \lambda^2 - 2\lambda + 3\lambda - 6 - 4x_2$$

$$= \lambda^2 + \lambda - 6 - 4x_2$$

$$- \text{ in } \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \Rightarrow \lambda^2 + \lambda - 6 = 0 \Rightarrow \frac{-1 \pm \sqrt{1 - 4\cdot(-6)}}{2} \begin{cases} -\frac{6}{2} \\ \frac{4}{2} \end{cases}$$

$\Rightarrow \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$ instabile (un autovalore a parte reale positivo)

$$- \text{ in } \begin{Bmatrix} 6 \\ -3 \end{Bmatrix} \Rightarrow \lambda^2 + \lambda - 6 + 12 = \lambda + \lambda + 6 \Rightarrow \frac{-1 \pm \sqrt{-23}}{2}$$

$\Rightarrow$ parte reale negativa $\Rightarrow$ stato asintoticamente stabile

④

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t)$$

$$A = \begin{bmatrix} -1 & 3 \\ 0 & -2 \end{bmatrix} \qquad B = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \qquad C = [2 \quad 0]$$

- $u = 0$, calcolare $x(t)$ a $t_f = 5$ sec

ingresso nullo $\Rightarrow \begin{cases} \dot{x}(t) = Ax(t) \\ y(t) = Cx(t) \end{cases}$

$$\dot{x}(t) = e^{At}\cdot x_0$$

$$e^{At} = \mathcal{L}^{-1}[(sI - A)^{-1}] \qquad sI - A = \begin{bmatrix} \lambda+1 & -3 \\ 0 & \lambda+2 \end{bmatrix}$$

$$\det(sI - A) = (\lambda+1)(\lambda+2)$$

$$\text{adj}(sI - A)$$

$$C_{11} = 1\cdot\lambda+2 = \lambda+2$$

$$C_{12} = 0 \qquad \Rightarrow C = \begin{bmatrix} \lambda+2 & 0 \\ 3 & \lambda+1 \end{bmatrix} \qquad C^T = \begin{bmatrix} \lambda+2 & 3 \\ 0 & \lambda+1 \end{bmatrix}$$

$$C_{21} = -1\cdot(-3) = 3$$

$$C_{22} = 1\cdot\lambda+1 = \lambda+1 \qquad \frac{3}{(\lambda+1)(\lambda+2)}$$

$$(sI - A)^{-1} = \begin{bmatrix} \frac{\cancel{\lambda+2}}{(\lambda+1)\cancel{(\lambda+2)}} & \frac{3}{(\lambda+1)(\lambda+2)} \\ \cancel{\frac{0}{(\lambda+1)(\lambda+2)}} & \frac{\cancel{\lambda+1}}{\cancel{(\lambda+1)}(\lambda+2)} \end{bmatrix}$$

$$\mathcal{L}^{-1}\left[\frac{1}{\lambda+1}\right] = e^{-t} \qquad \mathcal{L}^{-1}\left[\frac{1}{\lambda+2}\right] = e^{-2t}$$

$$\frac{3}{(\lambda+1)(\lambda+2)} = \frac{A}{\lambda+1} + \frac{B}{\lambda+2} \Rightarrow A\lambda + 2A + B\lambda + B = 3$$

$$\Rightarrow \begin{cases} A + B = 0 \\ 2A + B = 3 \end{cases} \Rightarrow \begin{cases} A = -B \\ -2B + B = 3 \end{cases} \Rightarrow \begin{cases} A = 3 \\ B = -3 \end{cases}$$

$$\Rightarrow \frac{3}{\lambda+1} - \frac{3}{\lambda-1} \Rightarrow \mathcal{L}^{-1}\left[\frac{3}{\lambda+1} - \frac{3}{\lambda-1}\right] = 3e^{-t} - 3e^{t}$$

$$e^{At} = \begin{bmatrix} e^{-t} & 3e^{-t} - 3e^{t} \\ 0 & e^{-2t} \end{bmatrix}$$

$$y(t) = Ce^{At}x_0 = [2 \quad 0]\begin{bmatrix} e^{-t} & 3e^{-t} - 3e^{t} \\ 0 & e^{-2t} \end{bmatrix}\begin{Bmatrix} 2 \\ -1 \end{Bmatrix}$$

$$\Rightarrow 2e^{-t} - 3e^{-t} + 3e^{t} \xrightarrow{\text{in } t=5} y(5) = 2e^{-5} - 3e^{-5} + 3e^{5} = 3e^{5} - e^{-5}$$

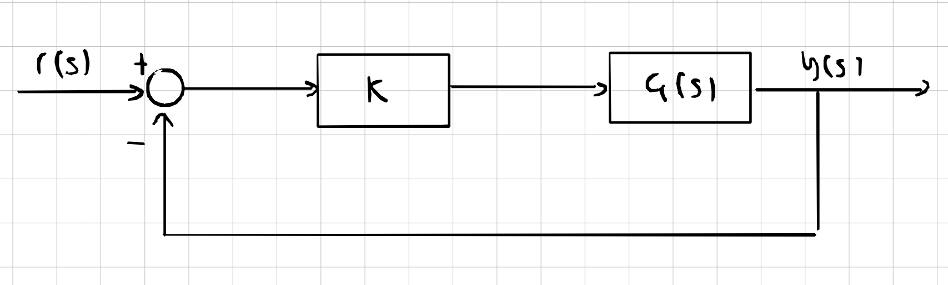
- $G(s)$

$$G(s) = C(sI - A)^{-1}B + \overset{0}{D} = [2 \quad 0]\begin{bmatrix} \frac{\cancel{\lambda+2}}{(\lambda+1)\cancel{(\lambda+2)}} & \frac{3}{(\lambda+1)(\lambda+2)} \\ \cancel{\frac{0}{(\lambda+1)(\lambda+2)}} & \frac{\cancel{\lambda+1}}{\cancel{(\lambda+1)}(\lambda+2)} \end{bmatrix}\begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$= -\frac{2}{(s+1)} + \frac{6}{(s+1)(s+2)}$$

$$\Rightarrow G(s) = \frac{-2(s+2) + 6}{(s+1)(s+2)} = \frac{-2s+2}{(s+1)(s+2)}$$

- Risposta armonica: –



$$G(s) = \frac{s+1}{s^2(s^2+4s+9)}$$

$$L(s) = K\cdot G = \frac{Ks+K}{s^2(s^2+4s+9)} \qquad 1 + L(s) = 0 \Rightarrow s^2(s^2+4s+9) + Ks + K = 0$$

$$\Rightarrow s^4 + 4s^3 + 9s^2 + Ks + K = 0$$

$$\begin{array}{c|ccc} 4 & 1 & 9 & K \\ 3 & 4 & K & 0 \\ 2 & -\frac{(K-36)}{4} & K & \\ 1 & \frac{20K-K^2}{-K+36} & 0 & \\ 0 & K & & \end{array}$$

$$X_{21} = -\frac{1}{4}\cdot\det\begin{vmatrix} 1 & 9 \\ 4 & K \end{vmatrix} = -\frac{(K-36)}{4}$$

$$X_{22} = -\frac{1}{4}\cdot\det\begin{vmatrix} 1 & K \\ 4 & 0 \end{vmatrix} = -\frac{1}{4}\cdot(-4K) = K$$

$$X_{11} = -\frac{4}{-K+36}\cdot\det\begin{vmatrix} 4 & K \\ -\frac{(K-36)}{4} & K \end{vmatrix}$$

$$= -\left(\frac{4}{-K+36}\right)\cdot\left(4K - \left(-\frac{(K-36)}{4}\cdot K\right)\right)$$

$$= -\left(\frac{\cancel{4}}{-K+36}\right)\cdot\left(\frac{16K + K^2 - 36K}{\cancel{4}}\right)$$

$$= -\left(\frac{K^2 - 20K}{-K+36}\right) = \frac{20K - K^2}{-K+36}$$

$$X_{01} = -\frac{(-K+36)}{20K - K^2}\cdot\det\begin{vmatrix} -\frac{(K-36)}{4} & K \\ \frac{20K-K^2}{-K+36} & 0 \end{vmatrix}$$

$$= +K$$

$$\Rightarrow \begin{cases} 36 - K > 0 \\ 20K - K^2 > 0 \\ K > 0 \end{cases}$$

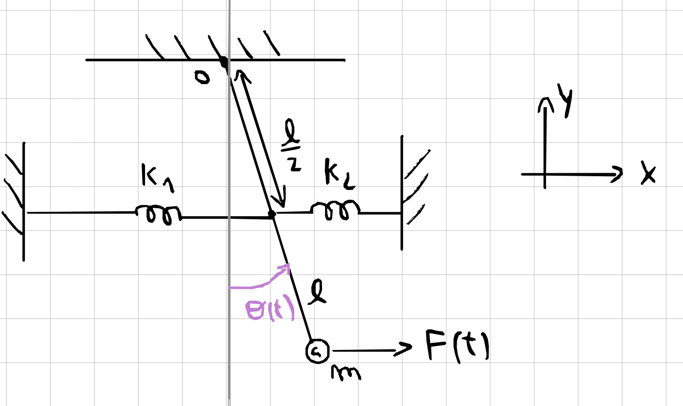
$$\Rightarrow K < 0 < 20$$

- Luogo delle radici: da capire se l'abbiamo fatto
- $r(s)\cdot L(s) = \frac{1}{s}\cdot L(s) \qquad y(\infty) = \lim_{s\to 0} \cancel{s}\cdot\frac{1}{\cancel{s}}L(s) =$

$$L(s) = \frac{KG(s)}{1 + KG(s)} = \frac{\frac{Ks+K}{s^2(s^2+4s+9)}}{1 + \frac{Ks+K}{s^2(s^2+4s+9)}} = \frac{\frac{Ks+K}{\cancel{s^2(s^2+4s+9)}}}{\frac{s^2(s^2+4s+9)+Ks+K}{\cancel{s^2(s^2+4s+9)}}}$$

$$= \frac{Ks+K}{s^2(s^2+4s+9) + Ks + K} \qquad y(\infty) = 1$$

Prova 6



$T = \frac{1}{2} m v^2$ $\quad OG = < l \cdot \sin\theta, -l \cdot \cos\theta >$

$\bar{v} = \dot{OG} = < + l \cos\theta\, \dot\theta, + l \sin\theta\, \dot\theta >$

$|\bar{v}|^2 = \left(\sqrt{l^2 \cos^2\theta\, \dot\theta^2 + l^2 \sin^2\theta\, \dot\theta^2}\right)^2 = l^2 \dot\theta^2 \quad \Rightarrow T = \frac{1}{2} m l^2 \dot\theta^2$

$U = U_P + U_E \quad U_P = -mg \cdot l \cos\theta \quad U_{E_1} = \frac{1}{2} k_1 \cdot \left(\frac{l}{2}\sin\theta\right)^2 = \frac{1}{2} k_1 \cdot \frac{l^2}{4} \sin^2\theta$

$= \frac{k_1 l^2}{8} \sin^2\theta \quad U_{E_2} = \frac{1}{2} k_2 \cdot \left(\frac{l}{2} \sin\theta\right)^2 = \frac{1}{2} k_2 \cdot \frac{l^2}{4} \sin^2\theta = \frac{k_2 l^2}{8} \sin^2\theta$

$U_{TOT} = -mgl\cos\theta + \frac{k_1 l^2}{8} \sin^2\theta + \frac{k_2 l^2}{8} \sin^2\theta$

$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot\theta}\right) = \frac{d}{dt}\left(m l^2 \dot\theta\right) = m l^2 \ddot\theta \quad \frac{\partial T}{\partial\theta} = 0$

$\frac{\partial U}{\partial \theta} = mgl\sin\theta + \frac{k_1 l^2}{\cancel{8}4} \cdot \cancel{2} \sin\theta\cos\theta + \frac{k_2 l^2}{\cancel{8}4} \cdot \cancel{2} \sin\theta\cos\theta$

$\delta L = F \cdot \delta x = F \cdot \delta(l \sin\theta) = F \cdot \frac{d(l\sin\theta)}{d\theta} \delta\theta = F \cdot l \cos\theta\, \delta\theta = Q_1 \cdot \delta\theta$

$\Rightarrow Q_1 = F \cdot l \cdot \cos\theta$

$\Rightarrow m l^2 \ddot\theta + mgl\sin\theta + \frac{k_1 l^2}{4} \sin\theta\cos\theta + \frac{k_2 l^2}{4} \sin\theta\cos\theta = F \cdot l \cdot \cos\theta$

$\ddot\theta = \frac{F \cdot \cancel{l} \cos\theta}{m l^{\cancel{2}}} - \frac{\cancel{m}g\cancel{l}}{\cancel{m} l^{\cancel{2}}} \sin\theta - \frac{k_1}{4} \frac{\cancel{l^2}}{m\cancel{l^2}} \sin\theta\cos\theta - \frac{k_2 \cancel{l^2}}{4 m \cancel{l^2}} \sin\theta\cos\theta$

Scegliendo $x_1 = \theta_1 \quad x_2 = \dot\theta_1 \quad u -$

$$\begin{cases} \dot x_1 = x_2 \\ \dot x_2 = \frac{1}{ml} \cdot u \cdot \cos x_1 - \frac{g}{l} \sin x_1 - \frac{k_1}{4m} \sin x_1 \cos x_1 - \frac{k_2}{4m} \sin x_1 \cos x_1 \end{cases}$$

②

$$\begin{cases} \dot x_1(t) = (u(t)+1) \sin x_1(t) \\ \dot x_2(t) = x_1(t) + x_2(t) + 2u(t) \end{cases}$$

$y(t) = x_1(t) - 3x_2^2(t) + 3u(t)$

- $u = 0 \Rightarrow \begin{cases} \dot x_1 = \sin x_1 \\ \dot x_2 = x_1 + x_2 \end{cases} \quad \begin{cases} \dot x_1 = 0 \\ \dot x_2 = 0 \end{cases} \Rightarrow \begin{cases} \sin x_1 = 0 \\ x_2 = -x_1 \end{cases}$

$\bar{x}_e' = [\,0 \quad 0\,]^T \quad \bar{x}_e'' = [\,\pi \quad -\pi\,]^T$

$\frac{\partial f}{\partial x_1} = (\underset{0}{u(t)} + 1) \cdot \cos x_1 \quad \frac{\partial f}{\partial x_2} = 0$

$\frac{\partial f_2}{\partial x_1} = 1 \quad \frac{\partial f_2}{\partial x_2} = 1$

$\Rightarrow A = \begin{bmatrix} \cos x_1 & 0 \\ 1 & 1 \end{bmatrix}$

$B = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

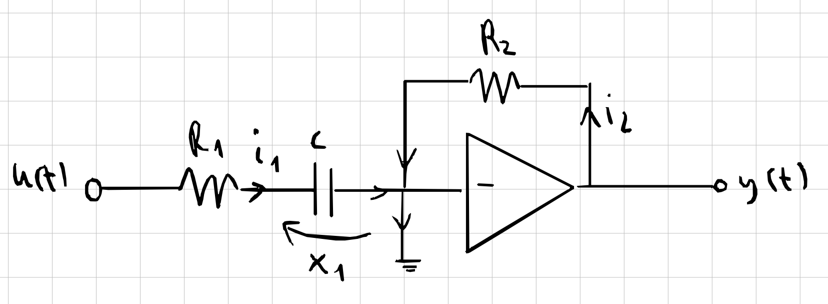
in $\begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

$C = [\,1 \quad -3\,] \quad D = [\,0\,]$

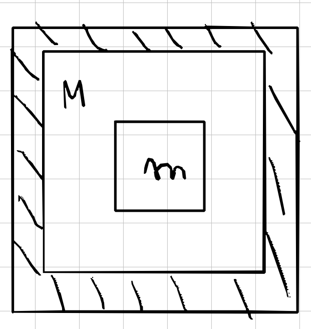
$\Rightarrow \lambda I - A = \begin{bmatrix} \lambda-1 & 0 \\ -1 & \lambda-1 \end{bmatrix}$

$\Rightarrow (\lambda-1)(\lambda-1)+1 = \lambda^2 - \lambda - \lambda + 1 + 1 = \lambda^2 - 2\lambda + 2$ $\Rightarrow$ instabile (grado 2 e coefficienti diversi tra loro)

③



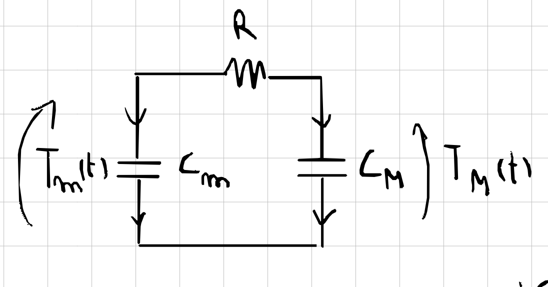
④



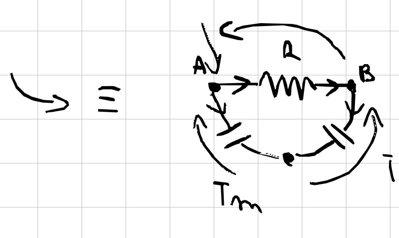
$M = 1\,\text{kg} \quad m = 0.5\,\text{kg}$

$c_M = 2 \cdot J/kg\,°C \quad c_m = 1\, J/kg\,°C$

$R = 0.5\, °C/W$



$x_1 = T_m \quad x_2 = T_M$



$LKC_A: \; C_m \cdot \dot x_1 + \frac{(x_1 - x_2)}{R} = 0$

$\Rightarrow C_m \dot x_1 = -\frac{(x_1 - x_2)}{R}$

$\frac{(x_1 - x_2)}{R} - C_m \dot x_2 = 0$

$\Rightarrow C_m \dot x_2 = \frac{(x_1 - x_2)}{R}$

$\Rightarrow \begin{cases} \dot x_1 = -\frac{(x_1 - x_2)}{RC_m} \\ \dot x_2 = \frac{(x_1 - x_2)}{RC_M} \end{cases} \Rightarrow \begin{cases} \dot x_1 = -\frac{(x_1 - x_2)}{0.5} \\ \dot x_2 = \frac{(x_1 - x_2)}{1} \end{cases}$

$A = \begin{bmatrix} -2 & +2 \\ 1 & -1 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\lambda I - A = \begin{bmatrix} \lambda+2 & -2 \\ -1 & \lambda+1 \end{bmatrix} = (\lambda+2)(\lambda+1) - 2 = \lambda^2 + \lambda + 2\lambda + 3 - 2 = \lambda^2 + 3\lambda + 1$

$\frac{-3 \pm \sqrt{9-4}}{2} \sim \begin{cases} \frac{-3-\sqrt5}{2} \\ \frac{-3+\sqrt5}{2} \end{cases}$ $\quad e^{-\frac{3-\sqrt5}{2}}, \; e^{-\frac{3+\sqrt5}{2}}$ modi del sistema

Assumendo $T_m(0) = 20°C$ e $T_M(0) = 100°C$

$x(t) = e^{At} x$

⑤

$\ddot y(t) - y(t) = 5 \cdot u(t) \quad y(0) = 0 \quad \dot y(0) = 1 \quad u(t) = e^{-2t}$ a $t=0$

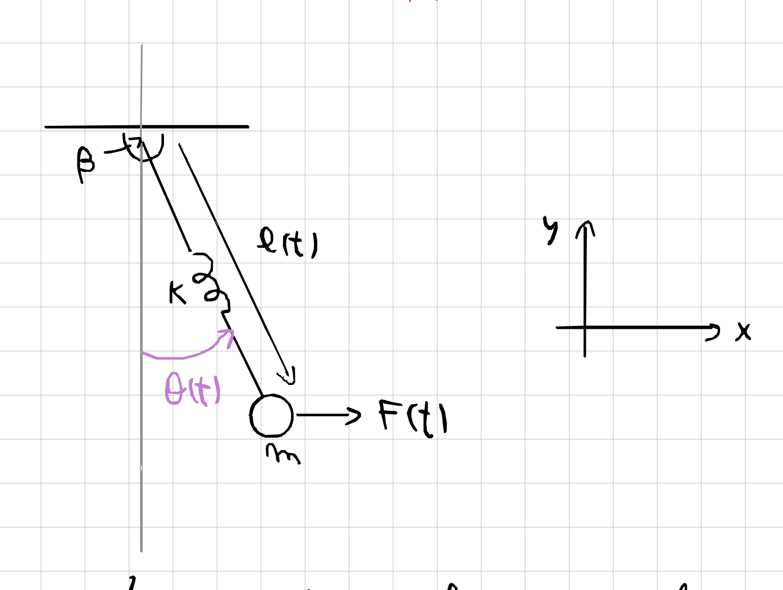
$\Rightarrow \ddot y(t) - y(t) = 5 \cdot e^{-2t}$ $\quad \downarrow \; \frac{1}{s+2}$

$\ddot y(t) - a_1 \dot y(t) - a_0 y(t)$

Prova 7

NO SOLUZIONI

①



$T = \frac{1}{2} m v^2$ $\quad OG = < l \cdot \sin\theta , - l\cos\theta >$

$\bar{v} = \dot{OG} = < \dot{l}\sin\theta + l\dot\theta\cos\theta , - \dot{l}\cos\theta + l\dot\theta\sin\theta >$

$|\bar{v}|^2 = (\underline{\dot{l}^2 \sin^2\theta} + l^2\dot\theta^2\cos^2\theta + \cancel{2\dot{l}l\dot\theta\sin\theta\cos\theta} + \underline{\dot{l}^2\cos^2\theta} + l^2\dot\theta^2\sin^2\theta - \cancel{2\dot{l}l\dot\theta\cos\theta\sin\theta})$

$= (\dot{l}^2 + l^2\dot\theta^2)$

$\Rightarrow T = \frac{1}{2} m (\dot{l}^2 + l^2\dot\theta^2)$

$U = U_P + U_E$ $\quad U_P = - mg(-l\cos\theta) = mgl\cos\theta$

$U_E = \frac{1}{2} K x^2 = \frac{1}{2} \cdot K \cdot (l - l_0)^2$

$U_{TOT} = mgl\cos\theta + \frac{1}{2} K (l - l_0)^2$

$R = \frac{1}{2} \beta \cdot \dot\theta^2$

$\frac{d}{dt}\left(\frac{\partial T}{\partial\dot\theta}\right) = \frac{d}{dt}\left(ml^2\dot\theta\right) = ml^2\ddot\theta$ $\quad \frac{\partial T}{\partial\theta} = 0$ $\quad \frac{\partial U}{\partial\theta} = - mgl\sin\theta$

$\frac{\partial R}{\partial\dot\theta} = \beta\dot\theta$

$\frac{d}{dt}\left(\frac{\partial T}{\partial\dot{l}}\right) = \frac{d}{dt}\left(m\dot{l}\right) = m\ddot{l}$ $\quad \boxed{\frac{\partial T}{\partial l} = ml\dot\theta^2}$ $\quad \frac{\partial U}{\partial l} = mg\cos\theta + \frac{1}{\cancel{2}}K\cdot\cancel{2}(l-l_0)$

$\frac{\partial R}{\partial\dot{l}} = 0$

$\delta L = F \cdot \delta(l \cdot \sin\theta) = F \cdot \sin\theta \cdot \delta l + F \cdot l \cdot \cos\theta \, \delta\theta$

Scegliendo $q_1 = l$ e $q_2 = \theta$ $\Rightarrow$ $Q_1 = F \cdot \sin\theta$ $\quad Q_2 = F \cdot l \cdot \cos\theta$

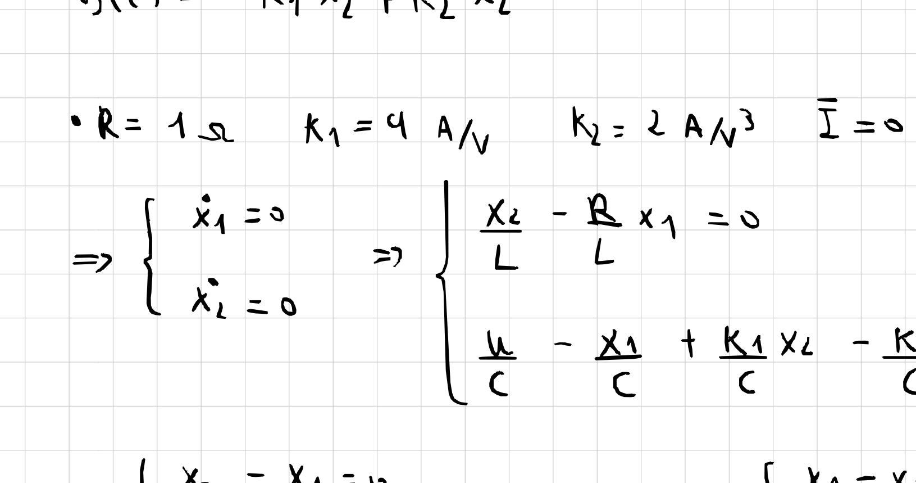
$$\begin{cases} m\ddot{l} - ml\dot\theta^2 + mg\cos\theta + K(l-l_0) = F\cdot\sin\theta \\ ml^2\ddot\theta - mgl\sin\theta + \beta\dot\theta^2 = F\cdot l\cdot\cos\theta \end{cases}$$

$$\begin{cases} \ddot{l} = \frac{F\cdot\sin\theta}{m} + l\dot\theta^2 - g\cdot\cos\theta - \frac{K(l-l_0)}{m} \\ \ddot\theta = \frac{F\cdot\cancel{l}\cdot\cos\theta}{ml^{\cancel{2}}} + \frac{\cancel{m}g\cancel{l}\sin\theta}{\cancel{m}l^{\cancel{2}}} - \frac{\beta\dot\theta^2}{ml^2} \end{cases}$$

$x_1 = l \quad x_2 = \dot{l} \quad x_3 = \theta \quad x_4 = \dot\theta \qquad u_1 = F \quad u_2 = l_0$

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = \frac{1}{m} \cdot u_1\sin x_3 + l \cdot x_4^2 - g \cdot \cos x_3 - \frac{K\cdot(x_1 - u_2)}{m} \\ \dot{x}_3 = x_4 \\ \dot{x}_4 = \frac{1}{m} \cdot u_1 \cos x_3 + \frac{g}{x_1} \cdot \sin x_3 - \frac{\beta}{m} \cdot \frac{x_4^2}{x_1} \end{cases}$$

②



$i(t) = -K_1 v(t) + K_2 v^3(t)$

$K_1 > 0$ e $K_2 > 0$

$x_2 = L\dot{x}_1 + R \cdot x_1 = v(t)$ $\quad \Rightarrow i(t) = - K_1 \cdot x_2 + K_2 x_2^3$

$LKC_A : I - x_1 - C\dot{x}_2 - i(t) = 0 \quad \Rightarrow I - x_1 - C\dot{x}_2 + K_1 x_2 - K_2 x_2^3 = 0$

$L\dot{x}_1 = x_2 - Rx_1$

$u = I(t)$

$$\Rightarrow \begin{cases} \dot{x}_1 = \frac{x_2}{L} - \frac{R}{L}x_1 \\ \dot{x}_2 = \frac{u}{C} - \frac{x_1}{C} + \frac{K_1}{C}x_2 - \frac{K_2}{C}x_2^3 \end{cases}$$

$y(t) = - K_1 x_2 + K_2 x_2^3$

• $R = 1\,\Omega \quad K_1 = 4\ A/V \quad K_2 = 2\ A/V^3 \quad \bar{I} = 0$

$$\Rightarrow \begin{cases} \dot{x}_1 = 0 \\ \dot{x}_2 = 0 \end{cases} \Rightarrow \begin{cases} \frac{x_2}{L} - \frac{R}{L}x_1 = 0 \\ \frac{u}{C} - \frac{x_1}{C} + \frac{K_1}{C}x_2 - \frac{K_2}{C}x_2^3 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} x_2 - x_1 = 0 \\ -x_1 + 4x_2 - 2x_2^3 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = x_2 \\ 4x_2 - x_2 - 2x_2^3 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} x_1 = x_2 \\ \cancel{2}x_2^3 - \overset{4}{\cancel{8}}x_2 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = x_2 \\ x_2(x_2^2 - 4) = 0 \end{cases}$$

$\bar{x}'_e = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad \bar{x}''_e = \begin{Bmatrix} 2 \\ 2 \end{Bmatrix} \quad \bar{x}'''_e = \begin{Bmatrix} -2 \\ -2 \end{Bmatrix}$

$f_1 = x_2 - x_1$

$f_2 = -x_1 + 4x_2 - 2x_2^3$

$\frac{\partial f_1}{\partial x_1} = -1 \quad \frac{\partial f_1}{\partial x_2} = 1 \quad \frac{\partial f_2}{\partial x_1} = -1 \quad \frac{\partial f_2}{\partial x_2} = 4 - 2x_2^3$

$\Rightarrow A = \begin{bmatrix} -1 & 1 \\ -1 & 4-2x_2^3 \end{bmatrix}$ in $0,0$ $\Rightarrow A = \begin{bmatrix} -1 & 1 \\ -1 & 4 \end{bmatrix}$ $\quad B = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$y(t) = - \underset{\substack{\downarrow \\ 4}}{K_1} \overset{\substack{0 \\ \uparrow}}{x_2} + \underset{\substack{\downarrow \\ 2}}{K_2} \overset{\substack{0 \\ \uparrow}}{x_2^3} = - K_1 + 3K_2 x_2^2 \quad \Rightarrow \quad C = [\,0 \quad 0\,] \quad D = [\,0\,]$

↑ capire se ha senso

$\lambda I - A = \begin{bmatrix} \lambda+1 & -1 \\ 1 & \lambda-4+2x_2^3 \end{bmatrix}$ $\quad \det(\lambda I - A) = (\lambda+1)(\lambda - 4 + 2x_2^3)$

$= \lambda^2 - 4\lambda + 2x_2^3\lambda + \lambda - 4 + 2x_2^3$

$= \lambda^2 + (2x_2^3 - 8)\lambda - 4 + 2x_2^3$

in $\begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$ $\Rightarrow \lambda^2 - 8\lambda - 4$ $\Rightarrow$ coefficienti di segno diverso $\Rightarrow$ instabile (e grado 2)

in $\begin{Bmatrix} 2 \\ 2 \end{Bmatrix}$ $\Rightarrow \lambda^2 + 8\lambda + 7$ $\Rightarrow$ coefficienti di segno uguale $\Rightarrow$ asintoticamente stabile (e grado 2)

in $\begin{Bmatrix} -2 \\ -2 \end{Bmatrix}$ $\Rightarrow \lambda^2 - 24\lambda - 25$ $\Rightarrow$ coefficienti di segno diverso $\Rightarrow$ instabile (e grado 2)

③ $\dot{x}(t) = Ax(t) + Bu(t)$ $\quad A = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad C = [\,1 \quad 0\,]$

$y(t) = Cx(t)$

• $x(t) = e^{At} \cdot x_0$

$e^{At} = \mathcal{L}^{-1}[(sI - A)^{-1}]$ $\quad sI - A = \begin{bmatrix} \lambda+2 & -1 \\ -1 & \lambda+2 \end{bmatrix}$

$\det(sI - A) = (\lambda+2)(\lambda+2) - 1 = \lambda^2 + 2\lambda + 2\lambda + 4 - 1 = \lambda^2 + 4\lambda + 3$

$-2 \pm \sqrt{4-3} = -2 \pm 1 \begin{cases} -3 \\ -1 \end{cases} \quad \Rightarrow (\lambda-3)(\lambda-1)$

adj $(sI - A)$:

$C_{11} = \lambda + 2$

$C_{12} = -1 \cdot (-1)$ $\quad \Rightarrow C = \begin{bmatrix} \lambda+2 & 1 \\ 1 & \lambda+2 \end{bmatrix}$ $\quad C^T = C$

$C_{21} = -1\ (-1)$

$C_{22} = \lambda + 2$

$$\Rightarrow (sI - A)^{-1} = \begin{bmatrix} \frac{\lambda+2}{(\lambda-3)(\lambda-1)} & \frac{1}{(\lambda-3)(\lambda-1)} \\ \frac{1}{(\lambda-3)(\lambda-1)} & \frac{\lambda+2}{(\lambda-3)(\lambda-1)} \end{bmatrix}$$

$\frac{\lambda+2}{(\lambda-3)(\lambda-1)} = \frac{A}{\lambda-3} + \frac{B}{\lambda-1} \quad \Rightarrow A\lambda - A + B\lambda - 3B = \lambda + 2$

$$\Rightarrow \begin{cases} A + B = 1 \\ -A - 3B = 2 \end{cases} \Rightarrow \begin{cases} A = 1 - B \\ -1 + B - 3B = 2 \end{cases} \Rightarrow \begin{cases} '' \\ -2B = 3 \end{cases} \Rightarrow \begin{cases} A = \frac{1}{3} \\ B = \frac{2}{3} \end{cases}$$

$\Rightarrow \frac{1}{3}\left(\frac{1}{\lambda-3}\right) + \frac{2}{3}\left(\frac{1}{\lambda-1}\right)$ $\quad \mathcal{L}^{-1} = \frac{1}{3} \cdot e^{3t} + \frac{2}{3}e^t$

$\frac{1}{(\lambda-3)(\lambda-1)} = \frac{A}{\lambda-3} + \frac{B}{\lambda-1} \quad \Rightarrow A\lambda - A + B\lambda - 3B = 1$

$$\Rightarrow \begin{cases} A + B = 0 \\ -A - 3B = 1 \end{cases} \Rightarrow \begin{cases} A = -B \\ B - 3B = 1 \end{cases} \Rightarrow \begin{cases} A = \frac{1}{2} \\ B = -\frac{1}{2} \end{cases}$$

$\Rightarrow \frac{1}{2}\left(\frac{1}{\lambda-3}\right) - \frac{1}{2}\left(\frac{1}{\lambda-1}\right)$ $\quad \mathcal{L}^{-1} = \frac{1}{2}e^{3t} - \frac{1}{2}e^t$

$$e^{At} = \begin{bmatrix} \frac{1}{3}e^{3t} + \frac{2}{3}e^t & \frac{1}{2}e^{3t} - \frac{1}{2}e^t \\ \frac{1}{2}e^{3t} - \frac{1}{2}e^t & \frac{1}{3}e^{3t} + \frac{2}{3}e^t \end{bmatrix}$$

$$x(t) = \begin{bmatrix} \frac{1}{3}e^{3t} + \frac{2}{3}e^t & \frac{1}{2}e^{3t} - \frac{1}{2}e^t \\ \frac{1}{2}e^{3t} - \frac{1}{2}e^t & \frac{1}{3}e^{3t} + \frac{2}{3}e^t \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

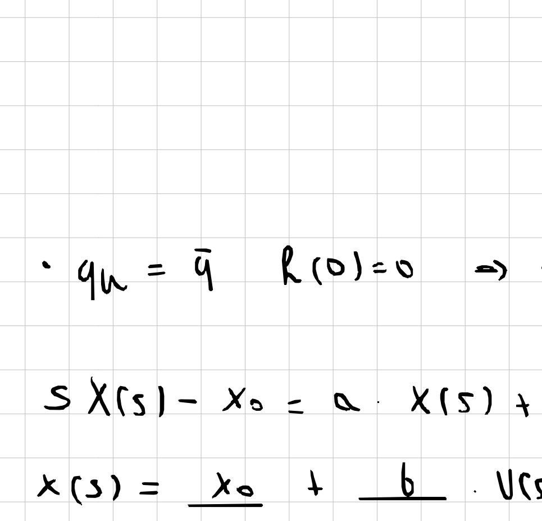
$$x(t) = \begin{bmatrix} e^{3t}\left(\frac{1}{3} + \frac{1}{2}\right) + e^t\left(\frac{2}{3} - \frac{1}{2}\right) \\ e^{3t}\left(\frac{5}{6}\right) + e^t\left(\frac{1}{6}\right) \end{bmatrix} = \begin{bmatrix} \frac{1}{6}e^t + \frac{5}{6}e^{3t} \\ \frac{1}{6}e^t + \frac{5}{6}e^{3t} \end{bmatrix}$$

$G(t) = C \cdot e^{At} \cdot B$ $\quad [\,1 \quad 0\,]\begin{bmatrix} \frac{1}{3}e^{3t} + \frac{2}{3}e^t & \frac{1}{2}e^{3t} - \frac{1}{2}e^t \\ \frac{1}{2}e^{3t} - \frac{1}{2}e^t & \frac{1}{3}e^{3t} + \frac{2}{3}e^t \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$= \frac{1}{3}e^{3t} + \frac{2}{3}e^t$

Risposta armonica: –

④



$A \cdot \dot{h} = q_u - q_y$

$P(t) - P_a = R \cdot q \quad \Rightarrow \rho g h = R \cdot q_y \Rightarrow q_y = \frac{\rho g h}{R}$

$\Rightarrow A \cdot \dot{h} = q_u - \frac{\rho g h}{R}$

$\Rightarrow \dot{h} = \frac{q_u}{A} - \frac{\rho g h}{AR}$

• $q_u = \bar{q} \quad h(0) = 0 \quad \Rightarrow \dot{h} = \frac{\bar{q}}{A} - \frac{\rho g h}{Ar}$ $\qquad A = \left[-\frac{\rho g}{AR}\right] = a$

$sX(s) - x_0 = a \cdot X(s) + b \cdot U(s)$ $\qquad B = \left[\frac{1}{A}\right] = b$

$x(s) = \frac{x_0}{s-a} + \frac{b}{s-a} \cdot U(s)$ $\qquad U(s) = \frac{\bar{q}}{A} \cdot \frac{1}{s}$

$\Rightarrow x(s) = \frac{\left(\frac{1}{A}\right)}{\left(s + \frac{\rho g}{AR}\right)} \cdot \frac{\bar{q}}{As} = \frac{q}{A^2 s} \cdot \frac{AR}{sAR + \rho g} = \frac{\bar{q}R}{s(sAR + \rho g)}$

$\Rightarrow \frac{K}{s} + \frac{B}{(sAR + \rho g)} \quad \Rightarrow KsAR + K\rho g + Bs = \bar{q}R$

$$\Rightarrow \begin{cases} KAR + B = 0 \\ +K\rho g = \bar{q}R \end{cases} \Rightarrow \begin{cases} B = -\frac{\bar{q}AR^2}{\rho g} \\ K = \frac{\bar{q}R}{\rho g} \end{cases}$$

$\Rightarrow \frac{\bar{q}R}{\rho g} \cdot \frac{1}{s} - \frac{\bar{q}AR^2}{\rho g} \cdot \frac{1}{sAR + \rho g} = \frac{\bar{q}R}{\rho g} \cdot \frac{1}{s} - \frac{\bar{q}\cancel{AR^2}}{\rho g} \cdot \frac{1}{\cancel{AR}\cdot\left(s + \frac{\rho g}{AR}\right)}$

$= \frac{q\bar{R}}{\rho g} - \frac{q\bar{R}}{\rho g} \cdot e^{-\frac{\rho g}{AR}t}$

$h(\infty) = \lim_{s\to 0} sX(s) = \lim_{s\to 0} \cancel{s}\frac{\bar{q}R}{\cancel{s}(sAR + \rho g)} = \frac{\bar{q}R}{\rho g}$

• $q_u(t) = 0 \quad h(0) = \frac{3R}{\rho g}$

$h(t) = e^{at} \cdot x_0 \quad \Rightarrow h(t) = e^{-\frac{\rho g}{AR}t} \cdot \frac{3R}{\rho g}$

$\frac{\cancel{R}}{\cancel{\rho g}} = e^{-\frac{\rho g}{AR}t} \cdot \frac{3\cancel{R}}{\cancel{\rho g}} = e^{-\frac{\rho g}{AR}t} = \frac{1}{3} \quad \Rightarrow -\frac{\rho g}{AR}t = \log\left(\frac{1}{3}\right)$

$\Rightarrow t = \frac{\log(1/3)}{\left(-\frac{\rho g}{AR}\right)}$

⑤ –

Prova 10/04/2014

1) $T_1 = \frac{1}{2} J \dot{\psi}^2$ $\quad T_2 = \frac{1}{2} M_1 \dot{s}_1^2$ $\quad T_3 = \frac{1}{2} M_2 \dot{s}_2^2$

$$\dot{\overline{OG}} = < l \sin\theta + s ,\ l \cos\theta > \qquad \bar{v} = \dot{\overline{OG}} = < \dot\theta\, l \cos\theta + \dot s ,\ - l \sin\theta >$$

$$|\bar v|^2 = \dot\theta^2 l^2 \cos^2\theta + \dot s_2^2 + 2 \dot s_2 \dot\theta\, l \cos\theta + l^2 \sin^2\dot\theta^2 = \dot\theta^2 l^2 + \dot s(\dot s + 2\dot\theta l \cos\theta)$$

$$\Rightarrow T_4 = \frac{1}{2} m (\dot\theta^2 l^2 + \dot s_2(\dot s_2 + 2\dot\theta l \cos\theta)) + \frac{1}{2} J_4 \dot\theta^2$$

$$T = \frac{1}{2} J \dot\psi^2 + \frac{1}{2} M_1 \dot s_1^2 + \frac{1}{2} M_2 \dot s_2^2 + \frac{1}{2} m (\dot\theta^2 l^2 + \dot s(\dot s + 2\dot\theta l \cos\theta)) + \frac{1}{2} J_4 \dot\theta^2$$

$$\psi \cdot r = s_1 \quad \Rightarrow \quad \psi = \frac{s_1}{r}$$

$$\Rightarrow T = \frac{1}{2}\frac{J}{r^2} \dot s_1^2 + \frac{1}{2} M_1 \dot s_1^2 + \frac{1}{2} M_2 \dot s_2^2 + \frac{1}{2} m (\dot\theta^2 l^2 + \dot s_2(\dot s_2 + 2\dot\theta l \cos\theta)) + \frac{1}{2} J_4 \dot\theta^2$$

$$U = U_p + U_E \qquad U_p = - m \cdot g \cdot (s_2 + l \cos\theta)$$

$$U_E = \frac{1}{2} K \cdot (s_1 - s_2)^2 \qquad R = \frac{1}{2} \beta (\dot s_1 - \dot s_2)^2$$

$$U_{TOT} = - mg (s_2 + l \cos\theta) + \frac{1}{2} K (s_1 - s_2)^2$$

$$\frac{\partial T}{\partial \dot s_1} = \frac{J}{r^2} \dot s_1 + M_1 \dot s_1 \qquad \frac{d}{dt}\left(\frac{\partial T}{\partial \dot s_1}\right) = \frac{J}{r^2} \ddot s_1 + M_1 \ddot s_1 \qquad \frac{\partial T}{\partial s_1} = 0$$

$$\frac{\partial T}{\partial \dot s_2} = M_2 \dot s_2 + m \dot s_2 + m \dot\theta l \cos\theta \qquad \frac{d}{dt}\left(\frac{\partial T}{\partial \dot s_2}\right) = M_2 \ddot s_2 + m \ddot s_2 + m \ddot\theta l \cos$$

$$\frac{\partial T}{\partial \dot\theta} = m \dot\theta l^2 + \frac{1}{2} m \dot s_2 l \cos\theta + J_4 \dot\theta \qquad \frac{d}{dt}\left(\frac{\partial T}{\partial \dot\theta}\right) = m \ddot\theta l^2 - \frac{1}{2} m \dot s_2 l \theta \sin\theta + J_4 \ddot\theta$$

$$\frac{\partial U_{TOT}}{\partial \theta} = - mg l \sin\theta \qquad \frac{\partial U_{TOT}}{\partial s_1} = K(s_1 - s_2)$$

$$\frac{\partial U_{TOT}}{\partial s_2} = - mg - K(s_1 - s_2)$$

$$\frac{\partial R}{\partial \theta} = 0 \qquad \frac{\partial R}{\partial s_1} = \beta(\dot s_1 - \dot s_2) \qquad \frac{\partial R}{\partial s_2} = -\beta(\dot s_1 - \dot s_2)$$

Forza esterna: $C \quad \Rightarrow \quad \delta L = F \cdot \delta r = C \cdot \delta\psi = C \cdot \delta \frac{s_1}{r}$

$$C \cdot \frac{d}{d\theta}\frac{s_1}{r}\delta\theta + C \cdot \frac{d}{ds_1}\frac{s_1}{r}\delta s_1 + C \cdot \frac{d}{ds_2}\frac{s_1}{r} \cdot \delta s_2 = 0 \cdot \delta\theta + \frac{C}{r}\delta s_1 + 0\, \delta s_2$$

$$Q_1 = \frac{C}{r} \qquad Q_2 = 0 \qquad Q_3 = 0$$

$$\Rightarrow \begin{cases} m \ddot\theta l^2 - \frac{1}{2} m \dot s_2 l \theta \sin\theta + J_4 \ddot\theta - mg l \sin\theta = \frac{C}{r} \\ \frac{J}{r^2} \ddot s_1 + M_1 \ddot s_1 + K(s_1 - s_2) + \beta(\dot s_1 - \dot s_2) = 0 \\ M_2 \ddot s_2 + m \ddot s_2 + m \ddot\theta l \cos\theta - mg - K(s_1 - s_2) - \beta(\dot s_1 - \dot s_2) = 0 \end{cases}$$

Sistemi idraulici

1) $q_1 - q_y = A\cdot\dot h$ $\qquad x_1 = h \quad x_2 = q_1 \quad x_3 = q_y$

$P_u - P = R\cdot q_1 + L\dot q_1$ $\qquad P = \rho\cdot g\cdot h$

$P - P_A = R\cdot q_y + L\cdot\dot q_y$

$$\begin{cases}\dot x_1 = \frac{x_2}{A} - \frac{x_3}{A}\\ u - \rho g h = R\cdot q_1 + L\cdot\dot q_1\\ \rho g h = R\cdot q_y + L\cdot\dot q_y\end{cases} \Rightarrow \begin{cases}\dot x_1 = \frac{x_2}{A} - \frac{x_3}{A}\\ \dot x_2 = \frac{u}{L} - \frac{\rho g}{L}x_1 - \frac{Rx_2}{L}\\ \dot x_3 = \frac{\rho g}{L}x_1 - \frac{R}{L}\cdot x_3\end{cases}$$

$$A = \begin{bmatrix} 0 & \frac{1}{A} & -\frac{1}{A}\\ -\frac{\rho g}{L} & -\frac{R}{L} & 0\\ \frac{\rho g}{L} & 0 & -\frac{R}{L}\end{bmatrix} \quad B = \begin{bmatrix}0\\ \frac{1}{L}\\ 0\end{bmatrix} \quad C = \begin{bmatrix}1 & 0 & 0\\ 0 & 0 & 1\end{bmatrix} \quad D = \begin{bmatrix}0\\ 0\end{bmatrix}$$

$$\begin{cases}\frac{x_2}{\cancel{A}} - \frac{x_3}{\cancel{A}} = 0\\ \frac{2}{\cancel{L}} - \frac{\rho g}{\cancel{L}}x_1 - \frac{Rx_2}{\cancel{L}} = 0\\ \frac{\rho g}{\cancel{L}}x_1 - \frac{R}{\cancel{L}}\cdot x_3 = 0\end{cases} \quad \begin{cases}x_2 = x_3\\ -\rho g x_1 - Rx_3 = -2\\ x_1 = \frac{R}{\rho g}x_3\end{cases} \quad \begin{cases}x_2 = x_3\\ \cancel{2}Rx_3 = -\cancel{2}\\ //\end{cases}$$

$$\begin{cases}x_2 = x_3\\ x_3 = \frac{1}{R}\\ x_1 = \frac{1}{\rho g}\end{cases} \Rightarrow \begin{cases}\frac{1}{\rho g}\\ \frac{1}{R}\\ \frac{1}{R}\end{cases}$$

2) $x_1 = h_1 \quad x_2 = q_1 \quad x_3 = q_2 \quad x_4 = h_2$

$A_1\cdot\dot h_1 = q_u - q_1 - q_2$ $\qquad u_1 = q_u \quad w_2 = d$

$\rho g(h_1 - d) = R\cdot q_1 + L\dot q_1$

$\rho g h_1 - \rho g h_2 = R\cdot q_2 + L\cdot\dot q_2$

$A_2\cdot\dot h_2 = q_1 + q_2 - q_y$

$\rho g h_2 = R\cdot q_y \Rightarrow q_y = \frac{\rho g h_2}{R}$

$$\begin{cases}\dot x_1 = \frac{u_1}{A_1} - \frac{x_2}{A_1} - \frac{x_3}{A_1}\\ \dot x_2 = \frac{\rho g}{L}(x_1 - w_2) - \frac{R}{L}\cdot x_2\\ \dot x_3 = \frac{\rho g}{L}x_1 - \frac{\rho g}{L}x_4 - \frac{R}{L}x_3\\ \dot x_4 = \frac{x_2}{A_2} + \frac{x_3}{A_2} - \frac{\rho g}{RA_2}x_4\end{cases} \quad A = \begin{bmatrix}0 & -\frac{1}{A_1} & -\frac{1}{A_1} & 0\\ \frac{\rho g}{L} & -\frac{R}{L} & 0 & 0\\ \frac{\rho g}{L} & 0 & -\frac{R}{L} & -\frac{\rho g}{L}\\ 0 & \frac{1}{A_2} & \frac{1}{A_2} & -\frac{\rho g}{RA_2}\end{bmatrix} \quad B = \begin{bmatrix}\frac{1}{A_1} & 0\\ 0 & -\frac{\rho g}{L}\\ 0 & 0\\ 0 & 0\end{bmatrix}$$

$y_1 = h_1 \quad y_2 = h_2$

$y_3 = q_y = \frac{\rho g}{R}h_2$

$$C = \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 0 & 0 & 1\\ 0 & 0 & 0 & \frac{\rho g}{R}\end{bmatrix} \quad D = \begin{bmatrix}0 & 0\\ 0 & 0\\ 0 & 0\end{bmatrix}$$

3) $x_1 = q_1 \quad x_2 = h(t) \quad x_3 = s \quad x_4 = \dot s$

$P_i = \frac{F}{A_p}$ $\qquad \frac{F}{A_p} - \rho g h = R\cdot q_1 + L\cdot\dot q_1 + \rho g d$

$A_s\cdot\dot h = q_1 - q_2$

$q_2 = A_p\cdot\dot s$

$m\ddot s = F_u - Ks - \beta\dot s$

$\rho g h - \frac{F_u}{A_p} = R\cdot q_2 + L\dot q_2 \Rightarrow -F_u = -\rho g h A_p + RA_p q_2 + LA_p\dot q_2$

$m\ddot s = \rho g h A_p - RA_p q_2 - LA_p\dot q_2 - Ks - \beta\dot s = \rho g h\cdot A_p - RA_p^2\dot s - LA_p^2\ddot s - Ks - \beta\dot s$

$\Rightarrow \ddot s(m + LA_p^2) = \rho g h\cdot A_p - RA_p^2\dot s - Ks - \beta\dot s$

$$\begin{cases}\dot q_1 = \frac{F}{A_p\cdot L} - \frac{\rho g h}{L} - \frac{R}{L}q_1 - \frac{\rho g d}{L}\\ \dot h = \frac{q_1}{A_s} - \frac{A_p\cdot\dot s}{A_s}\\ \ddot s = \frac{\rho g h\cdot A_p}{(m + LA_p)} \frac{-RA_p^2\dot s}{(m + LA_p)} \frac{-Ks}{(m + LA_p^2)} \frac{-\beta\dot s}{(m + LA_p^2)}\\ \dot s = \dot s\end{cases}$$

$K_1 = m + La_p^2$

$$\Rightarrow \begin{cases}\dot x_1 = \frac{u_1}{A_pL} - \frac{\rho g}{L}x_2 - \frac{R}{L}x_1 - \frac{\rho g}{L}w_2\\ \dot x_2 = \frac{x_1}{A_s} - \frac{A_p}{A_s}\cdot x_4\\ \dot x_3 = x_4\\ \dot x_4 = \frac{\rho g A_p^2}{K_1}\cdot x_2 - \frac{RA_p^2}{K_1}x_4 - \frac{K}{K_1}x_3 - \frac{\beta}{K_1}x_4\end{cases}$$

$$A = \begin{bmatrix}-\frac{R}{L} & -\frac{\rho g}{L} & 0 & 0\\ \frac{1}{A_s} & 0 & 0 & -\frac{A_p}{A_s}\\ 0 & 0 & 0 & 1\\ 0 & \frac{\rho g A_p}{K_1} & -\frac{K}{K_1} & \frac{(-RA_p^2 - \beta)}{K_1}\end{bmatrix} \quad B = \begin{bmatrix}\frac{1}{A_pL} & -\frac{\rho g}{L}\\ 0 & 0\\ 0 & 0\\ 0 & 0\end{bmatrix}$$

$y_1 = s(t) \quad y_2 = h(t)$ $\qquad C = \begin{bmatrix}0 & 0 & 1 & 0\\ 0 & 1 & & 0\end{bmatrix} \quad D = \begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$

4) $x_1 = h_1 \quad x_2 : h_2 \quad x_3 = q_{23} \quad x_4 = h_3$

$A_1\cdot\dot h_1 = q_1 - q_{12}$ $\qquad \rho g h_1 = R_1 q_{12} \Rightarrow q_{12} = \frac{\rho g h_1}{R_1}$

$\Rightarrow A_1\cdot\dot h_1 = q_1 - \frac{\rho g}{R}h_1$

$A_2\,\dot h_2 = q_{12} - q_{22} - q_{23}$ $\qquad q_{22} = \frac{\rho g h_2}{R_3}$

$\rho g(h_2 - d_1) - \rho g h_3 = R_2\cdot q_{23} + L_2\,\dot q_{23} + \rho g(-d_2)$

$A_3\,\dot h_3 = q_{23} + q_2 - q_y$ $\qquad q_y = \frac{\rho g h_3}{R_4}$

$u_1 = q_1 \quad w_2 = q_2$

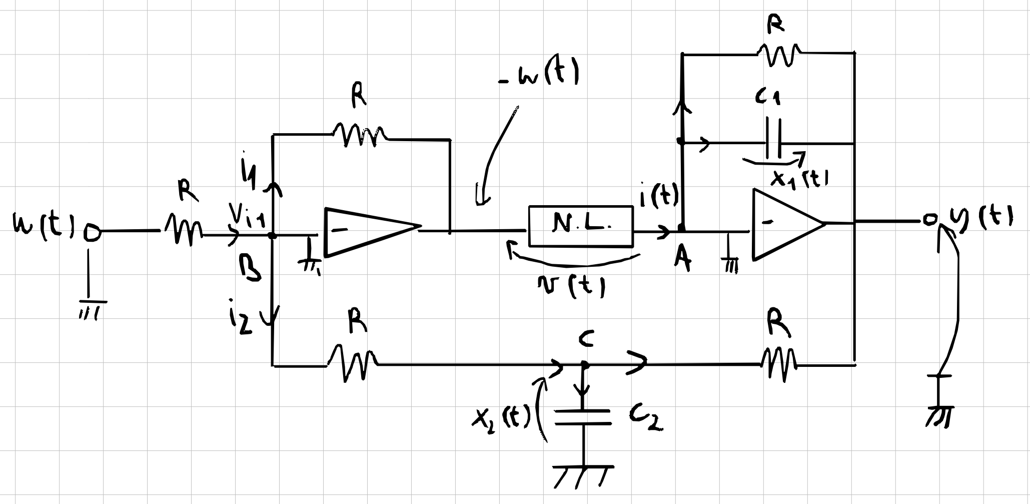
$$\begin{cases}A_1\cdot\dot x_1 = u_1 - \frac{\rho g}{R}x_1\\ A_2\cdot\dot x_2 = \frac{\rho g}{R_1}x_1 - \frac{\rho g}{R_3}x_2 - x_3\\ L\cdot\dot x_3 = \rho g x_2 - \rho g w_3 - \rho g x_4 - R_2\cdot x_3 + \rho g w_4\\ A_3\cdot\dot x_4 = x_3 + w_2 - \frac{\rho g}{R_4}\cdot x_4\end{cases}$$

$$\Rightarrow \begin{cases}\dot x_1 = \frac{w_1}{A_1} - \frac{\rho g}{A_1R}x_1\\ \dot x_2 = \frac{\rho g}{A_2R_1}x_1 - \frac{\rho g}{A_2R_3}x_2 - \frac{x_3}{A_2}\\ \dot x_3 = \frac{\rho g}{L}x_2 - \frac{\rho g}{L}w_3 - \frac{\rho g}{L}x_4 - \frac{R_2}{L}x_3 + \frac{\rho g}{L}w_4\\ \dot x_4 = \frac{x_3}{A_3} + \frac{w_2}{A_3} - \frac{\rho g}{A_3R_4}x_4\end{cases}$$

$$A = \begin{bmatrix}-\frac{\rho g}{A_1R_1} & 0 & 0 & 0\\ \frac{\rho g}{A_2R_1} & -\frac{\rho g}{A_2R_3} & -\frac{1}{A_2} & 0\\ 0 & \frac{\rho g}{L} & -\frac{R_2}{L} & -\frac{\rho g}{L}\\ 0 & 0 & \frac{1}{A_3} & -\frac{\rho g}{A_3R_4}\end{bmatrix} \quad B = \begin{bmatrix}\frac{1}{A_1} & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & -\frac{\rho g}{L} & \frac{\rho g}{L}\\ 0 & \frac{1}{A_3} & 0 & 0\end{bmatrix}$$

$$C = \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 0 & 0 & 1\\ 0 & 0 & 0 & \frac{\rho g}{R_4}\end{bmatrix} \quad D = \begin{bmatrix}0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\end{bmatrix}$$

Prove amplificatori



LKC$_A$: $i(t) + C\dot{x}_1 + R x_1 = 0$

LKC$_B$: $\frac{u}{R} - i_1 - i_2 = 0$

LKT$_1$: $R i_1 + v(t) = 0 \Rightarrow i_1 = -\frac{v(t)}{R}$

LKT : $x_2 + R i_2 = 0 \Rightarrow i_2 = -\frac{x_2}{R}$

$$\Rightarrow \frac{u}{\cancel{R}} + \frac{v(t)}{\cancel{R}} + \frac{x_2}{\cancel{R}} = 0 \Rightarrow v(t) = -(x_2 + u)$$

$$\Rightarrow i(t) = -k_1 \cdot (x_2 + u) + k_2 (x_2 + u)^2$$

$$\Rightarrow C\dot{x}_1 = k_1 (x_2 + u) - k_2 (x_2 + u)^2 - R x_1$$

$$\Rightarrow \dot{x}_1 = \frac{k_1}{C}(x_2 + u) - \frac{k_2}{C}(x_2 + u)^2 - \frac{R}{C} x_1$$

LKC$_C$: $i_2 - C_2 \dot{x}_2 - i_3 = 0$

LKT$_3$: $x_2 - i_3 \cdot R - x_1 = 0 \Rightarrow i_3 = \frac{x_2}{R} - \frac{x_1}{R}$

$$\Rightarrow C_2 \dot{x}_2 = -\frac{x_2}{R} + \frac{x_1}{R} - \frac{x_2}{R} \Rightarrow -\frac{2x_2}{R} + \frac{x_1}{R}$$

$$\Rightarrow \dot{x}_2 = \frac{-2x_2}{RC} + \frac{x_1}{RC}$$

$$\begin{cases} \dot{x}_1 = \frac{k_1}{C}(x_2 + u) - \frac{k_2}{C}(x_2 + u)^2 - \frac{1}{RC} x_1 \\ \dot{x}_2 = \frac{-2x_2}{RC} + \frac{x_1}{RC} \end{cases}$$

$R = 1 \quad k_1 = 4 \quad k_2 = 2$

$$\Rightarrow \begin{cases} \dot{x}_1 = \frac{4}{C}(x_2 + u) - \frac{2}{C}(x_2 + u)^2 - \frac{1}{C} x_1 \\ \dot{x}_2 = \frac{-2x_2}{RC} + \frac{x_1}{RC} \end{cases}$$

$$\Rightarrow \begin{cases} \dot{x}_1 = 0 \Rightarrow \frac{4}{\cancel{C}}(x_2 + \cancel{u}) - \frac{2}{\cancel{C}}(x_2 + \cancel{u})^2 - \frac{1}{\cancel{C}} x_1 = 0 \\ \dot{x}_2 = 0 \Rightarrow \frac{-2x_2}{\cancel{RC}} + \frac{x_1}{\cancel{RC}} = 0 \end{cases}$$

$$\Rightarrow \begin{cases} 4x_2 - 2x_2^2 - x_1 = 0 \\ x_1 = 2x_2 \end{cases} \Rightarrow \begin{cases} +\cancel{2}x^2 - \cancel{2}x_2 = 0 \\ x_1 = 2x_2 \end{cases} \Rightarrow \begin{matrix} x_e' = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \\ x_e'' = \begin{Bmatrix} 2 \\ 1 \end{Bmatrix} \end{matrix}$$

$$\frac{\partial f_1}{\partial x_1} = -1 \qquad \frac{\partial f_1}{\partial x_2} = 4 - 4x_2 \qquad \frac{\partial f_2}{\partial x_1} = 1 \qquad \frac{\partial f_2}{\partial x_2} = -2$$

$$\Rightarrow \begin{bmatrix} -1 & 4 - 4x_2 \\ 1 & -2 \end{bmatrix} \qquad \lambda_i - M = \begin{bmatrix} \lambda + 1 & 4x_2 - 4 \\ -1 & \lambda + 2 \end{bmatrix}$$

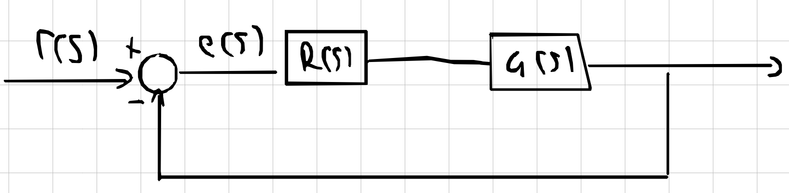
$$\Rightarrow (\lambda + 1)(\lambda + 2) + 4x_2 - 4 \Rightarrow \lambda^2 + 2\lambda + \lambda + 2 + 4x_2 - 4 = \lambda^2 + 3\lambda - 2 - 4x_2$$

in $\begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \Rightarrow \lambda^2 + 3\lambda - 2 = 0 \Rightarrow -3 \pm \sqrt{4 - 4 \cdot (-2)} \Rightarrow \frac{-3 \pm \sqrt{17}}{2}$

$\Rightarrow$ instabile

in $\begin{Bmatrix} 2 \\ 1 \end{Bmatrix} \Rightarrow \lambda^2 + 3\lambda - 0 \Rightarrow$ stabile $\Rightarrow$ segni concordi

Esercizi sistemi 1° e 2° ordine



$R = K_1 + \frac{K_2}{s}$ $\qquad G(s) = \frac{1}{s+4}$

$\hookrightarrow \frac{K_1 s + K_2}{s}$

$\Rightarrow G_R(s) = G \cdot R = \frac{K_1 s + K_2}{s(s+4)} \Rightarrow W(s) = \frac{\frac{K_1 s + K_2}{s(s+4)}}{1 + \frac{K_1 s + K_2}{s(s+4)}}$

$\Rightarrow \frac{\frac{K_1 s + K_2}{s\cancel{(s+4)}}}{\frac{s(s+4) + K_1 s + K_2}{s\cancel{(s+4)}}} \Rightarrow \frac{K_1 s + K_2}{s(s+4) + K_1 s + K_2}$

$\Rightarrow \frac{K_1 s + K_2}{s^2 + (4+K_1)s + K_2}$

$e(s) = r(s) - W \cdot r(s)$

$= \frac{1}{s^2} - W \cdot \frac{1}{s^2} = \frac{1}{s^2}\left(1 - W\right) \Rightarrow \frac{1}{s^2}\left(\frac{s(s+4) + \cancel{K_1 s} + \cancel{K_2} - \cancel{K_1 s} - \cancel{K_2}}{s(s+4) + K_1 s + K_2}\right)$

$\Rightarrow \frac{1}{s^2}\left(\frac{\cancel{s}(s+4)}{s(s+4) + K_1 s + K_2}\right) = \frac{1}{s}\left(\frac{s+4}{s(s+4) + K_1 s + K_2}\right)$

$e(\infty) = \lim_{t \to \infty} e(t) = \lim_{s \to 0} s \cdot e(s) = \lim_{s \to 0} \cancel{s} \cdot \frac{1}{\cancel{s}} \cdot \left(\frac{s+4}{s(s+4) + K_1 s + K_2}\right)$

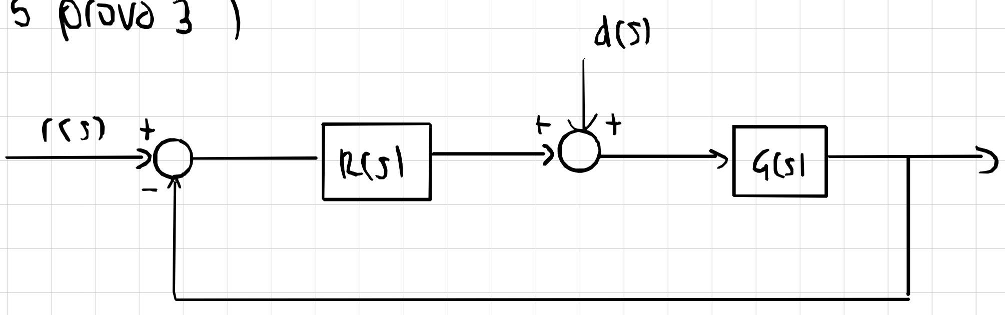
$= \frac{4}{K_2}$

L'errore a regime deve essere 0.01 $\Rightarrow \frac{4}{K_2} = 0.01 \Rightarrow K_2 = 400$

$W(s) = \frac{K_1 s + K_2}{s^2 + (4+K_1)s + K_2}$

$\begin{cases} 4 + K_1 = 2\delta\omega_n \\ \omega_n^2 = K_2 \end{cases} \Rightarrow \begin{cases} 4 + K_1 = \cancel{2} \cdot \frac{1}{\cancel{2}} \cdot 20 \\ \omega_n = \sqrt{K_2} = 20 \end{cases} \Rightarrow \begin{cases} K_1 = 16 \\ K_2 = 40 \end{cases}$

5 prova 3)



$R = \frac{K}{s+a}$ $\qquad G(s) = \frac{s+1}{s+2}$

$G_R(s) = R \cdot G = \frac{Ks + K}{(s+a)(s+2)} \Rightarrow W(s) = \frac{G_R}{1+G_R} = \frac{\frac{Ks+K}{(s+a)(s+2)}}{1 + \frac{Ks+K}{(s+a)(s+2)}}$

$\Rightarrow \frac{\frac{Ks+K}{\cancel{(s+a)(s+2)}}}{\frac{(s+a)(s+2) + Ks + K}{\cancel{(s+a)(s+2)}}} \Rightarrow \frac{Ks+K}{(s+a)(s+2) + Ks + K}$

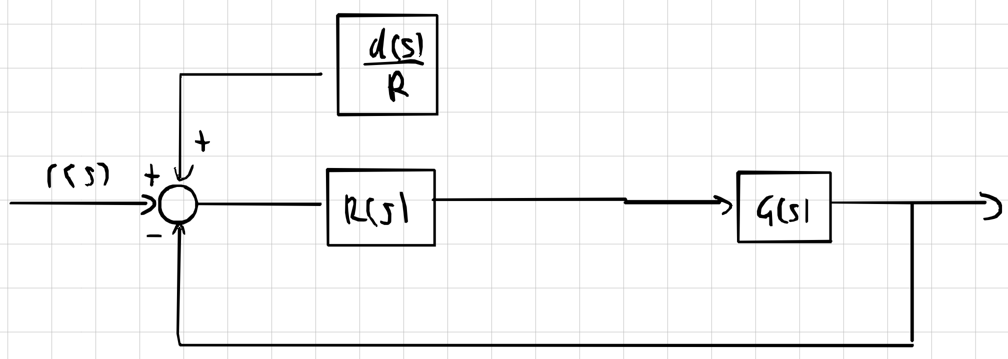
$\Rightarrow \frac{Ks+K}{s^2 + 2s + as + 2a + Ks + K} \Rightarrow \frac{Ks+K}{s^2 + (2+a+K)s + 2a + K}$

$T_{a5} = 0.2 \Rightarrow 0.2 = \frac{3}{\delta \cdot \omega_n} \Rightarrow \omega_n = 6 \Rightarrow 0.2 = \frac{\cancel{3}^{1}}{\delta \cdot \cancel{6}_{2}} = \delta = \frac{1}{2 \cdot 0.2}$

$\delta = \frac{5}{2}$

$\Rightarrow \delta = 10$

$\begin{cases} (2+a+K) = \cancel{2} \cdot \frac{5}{\cancel{2}} \cdot 6 \\ 2a + K = 36 \end{cases} \Rightarrow \begin{cases} 2+a+K = 30 \\ 2a + K = 36 \end{cases} \Rightarrow \begin{cases} K = 30 - a - 2 \\ 2a + \cancel{30} - a - 2 = \cancel{36}^{6} \end{cases} \Rightarrow \begin{cases} K = 20 \\ a = 8 \end{cases}$



$R(s) = \frac{20}{s+8}$ $\qquad G(s) = \frac{s+1}{s+2}$

$\frac{\left(\frac{1}{s}\right)}{\left(\frac{20}{s+8}\right)} \Rightarrow \frac{s+8}{20s}$ $\qquad G_R(s) = \frac{Ks+K}{(s+a)(s+2)}$ $\qquad W(s) = \frac{Ks+K}{s^2 + (2+a+K)s + 2a + K}$

$\Rightarrow W(s) = \frac{20s + 20}{s^2 + 30s + 36}$

$Y_s = W(s) \cdot \left(r(s) + \frac{d(s)}{R}\right)$

$Y_s = \frac{20s+20}{s^2 + 30s + 36} \cdot \left(\frac{1}{s} + \frac{(s+8)}{20s}\right) = \frac{20s+20}{s^2 + 30s + 36} \cdot \left(\frac{28+s}{20s}\right)$

$y(\infty) = \lim_{s \to 0} Y(s) = \cancel{s} \cdot \frac{\cancel{20s} + 20}{\cancel{s^2} + \cancel{30s} + 36} \cdot \left(\frac{28 + \cancel{s}}{20\cancel{s}}\right) \Rightarrow \frac{\cancel{20} \cdot \cancel{28}^{\,14\ 7}}{\cancel{20} \cdot 36_{\,18\ 4}} \Rightarrow \boxed{\frac{7}{4}}$

Esercizi risposta armonica

1) 2) $G(s) = C \cdot e^{At} \cdot B + D$